From Enumeration to Covering: Near-Optimal Densest $\mathcal{P}$-Partite Subgraph Search over Large Heterogeneous Information Networks

(Full Version)

Lu Chen, Chengfei Liu, Rui Zhou
Swinburne University of Technology
Melbourne, Australia
{luchen, cliu, rzhou}@swin.edu.au

Jiajie Xu
Soochow University
Suzhou, China
xujj@suda.edu.cn

Jianxin Li
Edith Cowan University
Perth, Australia
jianxin.li@ecu.edu.au

Abstract*—Given a heterogeneous information network (HIN) and a query meta-path $\mathcal{P}$ of length i, the densest $\mathcal{P}$-partite subgraph problem finds the subgraph, spanning the i typed layers of $\mathcal{P}$, that maximizes a parameter-free density: the number of meta-path instances over the geometric mean of the layer sizes. It has applications across bibliographic, e-commerce, and biomedical networks. The state-of-the-art approximation linearizes the geometric-mean objective by fixing per-layer weights, but solves one subproblem for every feasible weight set, of which there are $\mathcal{O}((n/i)^i)$, and on each achieves only a $1/i$ approximation. We show that neither the exhaustive enumeration nor the loose guarantee is necessary. First, we replace enumeration by covering: polylogarithmically many representative weight sets, localized further by a data-dependent bound, cover all feasible ones while losing only a tunable factor $1+\eta$ in density. Second, we cast each fixed-weight subproblem as a weighted supermodular densest-subgraph instance and solve it near-optimally, lifting the overall guarantee to $(1-\delta)/(1+\eta)$. To our knowledge, this is the first near-optimal density approximation beyond the bipartite (i=2) case, and it yields a PTAS for every fixed i. Algorithmically, our solver is an adaptive peeling scheme that never materializes the meta-path instances, whose number can exceed the graph size by orders of magnitude. Incumbent-driven pruning further discards representative weight sets before their subproblems are solved. Experiments on five real HINs show that our algorithms achieve substantial speedups over non-trivial baselines, and can further certify the near-optimality of the returned subgraph.

I. Introduction

Heterogeneous information networks. A heterogeneous information network (HIN) [1] carries a schema $T = (\mathcal{A}, \mathcal{R})$ of object types $\mathcal{A}$ and relation types $\mathcal{R}$: a graph $G = (V, E)$ with a vertex-type map $\phi : V \to \mathcal{A}$ and edge-type map $\psi : E \to \mathcal{R}$, $|\mathcal{A}| + |\mathcal{R}| > 2$. HIN can model bibliographic, e-commerce, knowledge-graph, and biomedical data such as HETIONET [2], which links compounds, genes, diseases, anatomies, and others (Figure 1). The schema lets one query at the type level through a *meta-path* [3]: a path $A_1 \xrightarrow{R_1} A_2 \xrightarrow{R_2} \cdots \xrightarrow{R_{i-1}} A_i$ on the schema T that alternates object types $A \in \mathcal{A}$ and relation types $R \in \mathcal{R}$, composing them into a single typed relation between its endpoints. When the connecting relations are understood, we abbreviate it by its *sequence of vertex types* $\mathcal{P} = \langle A_1, \ldots, A_i \rangle$; e.g. over HETIONET $\mathcal{P} = \langle \mathsf{C}, \mathsf{G}, \mathsf{D} \rangle$ (compound–gene–disease) links a drug to diseases via the genes it binds, of interest to drug repurposing.

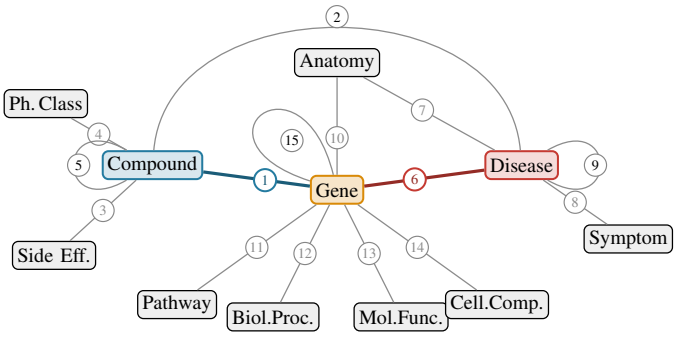
Cohesive $\mathcal{P}$-partite subgraphs. A $\mathcal{P}$-family is a tuple of vertex sets, one per type in $\mathcal{P}$; the subgraph it induces, the $\mathcal{P}$-partite graph, is typically large and sparse, while its meaningful information is hidden in the dense regions. Mining cohesion across multiple vertex sets has driven a long line of models, from multipartite cliques [4] and cross-graph quasi-cliques [5] to multilayer cores and densest subgraphs [6], [7], the relational community [8] and influential community [9]. Among these models, the densest $\mathcal{P}$-partite subgraph is especially appealing: a simple, parameter-free query surfacing cohesive structure across *every* type in $\mathcal{P}$, with no size, degree, or pattern threshold to tune. It stands to the pattern-based (k, P)-core [10] as the densest subgraph does to the k-core.

Densest $\mathcal{P}$-partite subgraph search (DPSS). Given a meta-path $\mathcal{P}$, DPSS [7] seeks a cohesive and semantically meaningful subgraph within the corresponding $\mathcal{P}$-partite graph. Its density is defined as the ratio between the number of $\mathcal{P}$-instances contained in the subgraph and the geometric mean of the sizes of its vertex sets across all vertex types. Mathematically, for a $\mathcal{P}$-family $\mathcal{V} = (V_1, \ldots, V_i)$ of i typed layers,

$$\rho(\mathcal{V}) = \frac{|\mathcal{F}(\mathcal{V})|}{\left(\prod_{j=1}^{i} |V_j|\right)^{1/i}}, \tag{1}$$

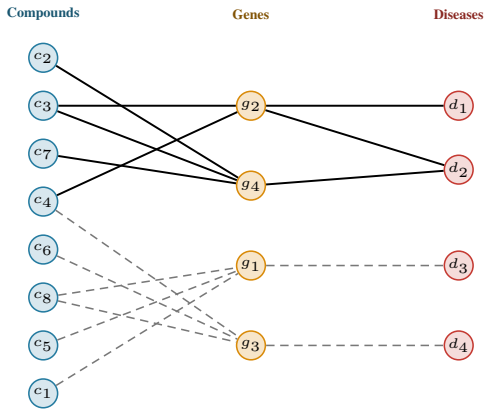
where $\mathcal{F}(\mathcal{V})$ is the set of $\mathcal{P}$-instances induced by $\mathcal{V}$, and DPSS returns the family of maximum density ρ^*. The geometric-mean normalization is what makes the objective parameter-free: it balances all i layers simultaneously, so that a high-density family must concentrate many meta-path instances among *relatively few* vertices of *every* type.

Data management significance. Modern database systems support multi-relational graphs for generality, like HINs. DPSS is a candidate *one-for-many* primitive for them for density based queries: one declarative query, parameterized only by $\mathcal{P}$, that specializes to the classical densest-subgraph families. Over



(1) compound **binds**/regulates a gene ; (2) treats/palliates a disease ; (3) causes a side effect ; (4) drug class includes a compound ; (5) resembles a compound ; (6) disease **associates**/regulates a gene ; (7) localizes to an anatomy ; (8) presents a symptom ; (9) resembles a disease ; (10) anatomy expresses/regulates a gene ; (11–14) gene participates in a pathway, process, etc. ; (15) gene interacts a gene.

Fig. 1. Schema of the Hetionet biomedical HIN



Compounds: c_1 masitinib, c_2 avapritinib, c_3 imatinib, c_4 dasatinib, c_5 crenolanib, c_6 axitinib, c_7 ripretinib, c_8 sunitinib. **Genes**: g_1 PDGFRA, g_2 ABL1, g_3 KDR, g_4 KIT. **Diseases**: d_1 chronic myeloid leukemia (CML), d_2 gastrointestinal stromal tumor (GIST), d_3 hypereosinophilic syndrome, d_4 renal cell carcinoma.

Fig. 2. A $\langle \mathsf{C}, \mathsf{G}, \mathsf{D} \rangle$-partite graph ($i$=3)

a single vertex type and its own relation, it is $|E|/|V|$, the densest subgraph; over two types ($i = 2$), it is the directed or bipartite densest subgraph; and for $i > 2$ it is the general meta-path density of Eq. (1), a $\mathcal{P}$-family dense across vertex types in $\mathcal{P}$.

Applications. As discussed, each densest family is a cohesive group spanning *every* type on the meta-path. Prior work motivates DPSS with such groups across HINs [7], exposing fraud rings over user–device–merchant networks, finding research communities over author–paper–venue networks, etc. We broaden the case to biomedical drug repurposing. Take $\mathcal{P} = \langle \mathsf{C}, \mathsf{G}, \mathsf{D} \rangle$ (compound–gene–disease) over a biomedical network such as HETIONET: a densest family is a set of compounds, genes, and diseases all densely interlinked across the three layers. In Figure 2, the solid $(4, 2, 2)$ block $\mathcal{V}^*$ is four kinase inhibitors acting through two genes (ABL1, KIT) on two cancers, CML and GIST. This block *suggests* drug-repurposing candidates: because *imatinib*, developed for CML via ABL1, shares the dense subgraph with GIST through KIT, the result identifies it as a candidate, a repurposing that in fact holds [11]. In contrast, edge-based methods are likely to miss it: the query places no edge between a compound and a disease; such a connection therefore tends to surface only through the *joint* density across all three vertex types.

State of the art. DPSS is polynomial-time solvable for fixed i, but only inefficiently, through costly flow computations; approximation is the more promising direction. The state-of-the-art approximation [7] fixes a per-layer (per-vertex type) weight set with product one, the *iRM-set*, to linearize the coupling geometric-mean denominator, turning DPSS into a weighted densest-subgraph subproblem. Two weaknesses remain: the optimal weight set is unknown, forcing a search over all of them, the set $\mathbb{M}$ of realizable iRM-sets, whose size can reach $\mathcal{O}((n/i)^i)$, and each is solved only to the loose $1/i$ factor of peeling. Both are already overcome at $i = 2$, the directed/bipartite densest subgraph [12], [13]: there, the weight set collapses to a single scalar ratio that is geometrically scanned, and each fixed-ratio subproblem is supermodular and solved near-optimally. This relies on one-dimensionality (the scalar ratio) and does not survive $i > 2$.

Open problems and challenges. The obstacle is twofold: for $i > 2$ the weight sets form a $(i-1)$-dimensional surface with *no* total order to sweep, and the numerator turns from the *edge* count into the *meta-path instance* count $|\mathcal{F}(\mathcal{V})|$, implicit and far larger than the graph size. This leads to three challenges. *(C1)* The enumeration of weight sets is $\mathcal{O}(|\mathbb{M}|)$, exponential in i; can approximation avoid it? *(C2)* The $1/i$ worst-case factor of peeling is often near-optimal in practice, yet never *certified* to be; can a fixed-weight subproblem, with its per-type-weighted denominator, be solved with a certified near-optimal guarantee? *(C3)* The materialized $\mathcal{P}$ instances number $\mathcal{O}(\prod_j |V_j|)$; can we avoid them?

Our approach. We attempt to address all three challenges. *(C1)* We replace enumeration with covering. In logarithmic coordinates, we flatten the feasibility $\prod_j m_j = 1$ to the hyperplane $\sum_j \ln m_j = 0$, where the feasible weights fill a bounded $(i-1)$-dimensional polytope; a single η-net covers it. We space that net at the coarse $\sqrt{\eta}$ pitch of the *second-order net*, which the curvature of the AM–GM gap justifies, keeping only $\mathcal{O}((\log n/\sqrt{\eta})^{i-1})$ representatives at a $1+\eta$ loss to the optimum density, $\eta^{-(i-1)/2}$ times fewer than the first-order $\ln(1+\eta)$ pitch, and we shrink $\log n$ to $\log \Delta$, where Δ is the largest per-vertex instance count, with a data-dependent bound. *(C2)* We cast each fixed-weight subproblem as a weighted supermodular densest-subgraph problem and solve it natively to a certified $(1-\delta)$ under its *per-type-weighted* denominator, the part the supermodular machinery, built for the cardinality ratio (the i=2 case), leaves out. That $(1-\delta)$ lives on the *weighted* objective, not the density; we prove it crosses over, since the AM–GM factor between them *vanishes at a family's own weights*, so at the optimum's weight our bound attains ρ^* itself. *(C3)* For meta-path instances, we factor each per-vertex incidence as $L(v)R(v)$, so an adaptive peel reads it with $\mathcal{O}(|E|)$ state and achieves the same $(1-\delta)$ *without ever materializing* the $\mathcal{O}(\prod_j |V_j|)$ instances. Together we get a $\frac{1-\delta}{1+\eta}$

approximation to ρ^*: a PTAS for fixed i and, to our knowledge, the first near-optimal guarantee past $i = 2$. Finally, we add incumbent-driven *pruning* that removes representatives before we solve them.

Contributions. In summary, this paper makes the following contributions.

- *Covering, not enumeration* (C1): instead of the $\mathcal{O}(|\mathbb{M}|)$ weight sets prior work enumerates, an η-net of only $\mathcal{O}((\min\{\log n, \log \Delta\}/\sqrt{\eta})^{i-1})$ representatives, polylogarithmic in n and Δ-adaptive, covers all of them at a $1+\eta$ loss (§V[1]).
- *Certified near-optimal subproblems* (C2): each fixed-weight step is a supermodular densest-subgraph instance, which we solve to a certified $(1-\delta)$ under its *per-type-weighted* denominator, the part the supermodular machinery leaves out; with the cover, this yields a $\frac{1-\delta}{1+\eta}$ PTAS for ρ^* at fixed i (§VI).
- *Instance-free peeling* (C3): for path meta-paths, an $L(v)R(v)$ factorization runs the peel with $\mathcal{O}(|E|)$ state, *without materializing* $\mathcal{P}$-instances (§VI-D).
- *Pruning*: an incumbent-driven test further removes representatives and reduces subproblem sizes (§VII).
- Experiments on real HINs show consistent gains in efficiency over existing methods with certified effectiveness (§VIII).

Omitted proofs are given in Appendices A and B.

II. Problem Formulation

In §I, we have defined the heterogeneous information network $G = (V, E)$, the query meta-path $\mathcal{P}$ of length $i = |\mathcal{P}|$, the $\mathcal{P}$-family $\mathcal{V} = (V_1, \ldots, V_i)$ and its induced $\mathcal{P}$-partite graph, the instance set $\mathcal{F}(\mathcal{V})$, and the density ρ. We do not restate them. We introduce the remaining notation and formally state the problem; throughout, $n = |V|$. Table I summarizes all symbols.

Typed vertices and positions. A meta-path may *repeat* a vertex type ($A_j = A_k$ for $j \neq k$), as in the gene–disease–gene path $\langle \mathsf{G}, \mathsf{D}, \mathsf{G} \rangle$, and its repeated positions are filled *independently*: a family may pick different sets $V_j \neq V_k$, of different sizes, for them. The type $\phi(v)$ then no longer identifies a vertex's role: when a type recurs, knowing $\phi(v)$ leaves it ambiguous which position v occupies. We therefore index by position and work over the *position-expanded* ground set $U = \biguplus_{j=1}^{i} V(A_j)$, one copy of $V(A_j)$ per position, so every $v \in U$ carries a definite position $\phi'(v) \in [i]$ (of required type $A_{\phi'(v)}$). A $\mathcal{P}$-family is then a subset of U, equivalently a tuple $\mathcal{V} = (V_1, \ldots, V_i)$ with $V_j \subseteq V(A_j)$, and a vertex whose type recurs appears once per position, chosen freely at each. Every per-position quantity is thereby unambiguous. When all types are distinct, ϕ' is fixed by ϕ and $U \cong V$.

Induced subgraph and per-vertex instances. A $\mathcal{P}$-family $\mathcal{V} = (V_1, \ldots, V_i)$ induces the subgraph $G(\mathcal{V})$ on $V_1 \cup \cdots \cup V_i$, and a $\mathcal{P}$-instance of $G(\mathcal{V})$ is a sequence $\pi = (v_1, \ldots, v_i)$ with $v_j \in V_j$ whose consecutive vertices are joined by the relation that $\mathcal{P}$ prescribes; we write $v \in \pi$ when v is one of its vertices.

[1]Throughout, § n abbreviates Section n.

TABLE I
Summary of notation.

Symbol	Meaning
$G = (V, E)$	input HIN (typed graph); $n = \lvert V \rvert$
$\mathcal{P},\ i$	meta-path; length $i = \lvert \mathcal{P} \rvert$
$A_j,\ V(A_j)$	position-j type; its vertex set
$U,\ \phi'(v)$	position-expanded $\biguplus_j V(A_j)$; position of v
$\phi(v)$	type of vertex v
$\mathcal{V}$	$\mathcal{P}$-family $(V_1, \ldots, V_i)$, $V_j \subseteq V(A_j)$
$G(\mathcal{V})$	subgraph induced by $\mathcal{V}$
$\mathcal{F}(\mathcal{V})$	$\mathcal{P}$-instances in $G(\mathcal{V})$
$P(v, G(\mathcal{V}))$	# $\mathcal{P}$-instances through v in $G(\mathcal{V})$
$g(\mathcal{V})$	$\left(\prod_j \lvert V_j \rvert\right)^{1/i}$
$\mathbf{m},\ m_j$	iRM-set; layer-j weight $g(\mathcal{V})/\lvert V_j \rvert$
Π_n	polytope of feasible iRM-sets (§V)
$\rho,\ \rho^*$	density; optimum density

By convention positions are matched independently, so a vertex may fill more than one role (i.e. $v_j = v_k$ is allowed when $A_j = A_k$) and every position-respecting sequence counts as a distinct instance. The algorithms that will be discussed operate not on $\mathcal{F}(\mathcal{V})$ directly but on its per-vertex incidence. As such, for a vertex v, we write $P(v, G(\mathcal{V})) = \left|\{\, \pi \in \mathcal{F}(\mathcal{V}) : v \in \pi \,\}\right|$ for the number of $\mathcal{P}$-instances of $G(\mathcal{V})$ passing through v. For brevity, we call it the incidence of v.

Example 1. Consider $\mathcal{V}^* = (\{c_2, c_3, c_4, c_7\}, \{g_2, g_4\}, \{d_1, d_2\})$, the solid edges of Figure 2: g_2 binds c_3, c_4 and reaches d_1, d_2, while g_4 binds c_2, c_3, c_7 and reaches d_2. Its $\mathcal{P}$-instances (the compound–gene–disease paths) number 7. The per-vertex incidences $P(v, G(\mathcal{V}^*))$ are $4, 3$ at g_2, g_4; $1, 3, 2, 1$ at c_2, c_3, c_4, c_7; and $2, 5$ at d_1, d_2.

Problem (Densest $\mathcal{P}$-Partite Subgraph Search, DPSS)**.** *Given a HIN $G = (V, E)$ and a meta-path $\mathcal{P}$, return a $\mathcal{P}$-family*

$$\mathcal{V}^* = \arg\max_{\mathcal{V}}\ \rho(\mathcal{V}), \qquad \rho^* = \rho(\mathcal{V}^*),$$

where $\rho(\mathcal{V}) = |\mathcal{F}(\mathcal{V})|/g(\mathcal{V})$ is the density of Eq. (1)*.*

Scope. DPSS is defined for every $i \geq 2$, but the case $i = 2$ is directed/bipartite densest subgraph, already solved near-optimally by existing methods [14], [15]. This paper targets the open general regime $i > 2$.

Example 2. The family $\mathcal{V}^*$ of Example 1 has density $\rho(\mathcal{V}^*) = \frac{7}{\sqrt[3]{4 \cdot 2 \cdot 2}} \approx 2.78$, and it is the densest $\mathcal{P}$-family in Figure 2, so $\rho^* \approx 2.78$. Enlarging it with the gray decoy ones (g_1, g_3 and their compounds c_1, c_5, c_6, c_8 and diseases d_3, d_4) gives the whole graph, of shape $(8, 4, 4)$ with 13 instances, yet its density is only $13/\sqrt[3]{128} \approx 2.58 < \rho^*$.

The iRM (ith-root-multiplication) set [7]. For a $\mathcal{P}$-family $\mathcal{V}$, the per-layer weights are $\mathbf{m} = (m_1, \ldots, m_i)$ with

$$m_j = \frac{g(\mathcal{V})}{|V_j|}, \qquad j \in [i],$$

rescaling each layer by its size. Fixing it turns DPSS into a tractable weighted subproblem (to be discussed in §III).

Property 1. *For every $\mathcal{P}$-family $\mathcal{V}$, the induced iRM-set satisfies $\prod_{j=1}^{i} m_j = 1$.*

iRM-set extension. We extend the notion beyond family-induced weights: *any* vector $\mathbf{m}$ with $\prod_j m_j = 1$ is a *feasible* iRM-set, and one actually induced by a $\mathcal{P}$-family is *realizable*.

III. State of the Art

We revisit the state of the art for DPSS and its $i = 2$ special case, isolating the gaps to fill.

The DPSS approximation. The geometric-mean density $\rho(\mathcal{V}) = |\mathcal{F}(\mathcal{V})|/g(\mathcal{V})$ resists direct optimization because its denominator $g = (\prod_j |V_j|)^{1/i}$ couples all i layers multiplicatively. Fixing a target iRM-set $\mathbf{m} = (m_1, \ldots, m_i)$ breaks the coupling and linearizes the denominator. Over the position-expanded ground set U of §II (with $S_j = S \cap V(A_j)$ for $S \subseteq U$ and position map ϕ'), fixing $\mathbf{m}$ replaces the geometric mean by the *weighted size*

$$w_{\mathbf{m}}(S) = \sum_{v \in S} m_{\phi'(v)} = \sum_j m_j |S_j|, \tag{2}$$

a weighted vertex count *linear* in S, equal to $i\,g(S)$ exactly when $\mathbf{m}$ is S's own iRM-set. The fixed-weight subproblem is then to maximize the *auxiliary objective*

$$h_{\mathbf{m}}(S) = \frac{f(S)}{w_{\mathbf{m}}(S)}, \qquad f(S) := |\mathcal{F}(S)|,$$

the instance count over a linear denominator, a *weighted densest-subgraph* problem, which [7] solves by *reweighted peeling*: it treats each vertex's incidence $P(v, G(\mathcal{V}))$ as its value and its weight $m_{\phi'(v)}$ as its cost, repeatedly deletes a vertex of least value-per-cost $P(v, G(\mathcal{V}))/m_{\phi'(v)}$, called its *key*, and keeps the densest family seen. A charging argument, each optimum instance charged to one of its i endpoints, bounds the loss for $\mathbf{m}$.

Theorem 1 ([7])**.** *For a fixed iRM-set $\mathbf{m}$, reweighted peeling is a $1/i$-approximation to the optimum density $\rho^*_{\mathbf{m}}$ over all $\mathcal{P}$-families conforming to $\mathbf{m}$ (i.e., whose own induced iRM-set equals the fixed $\mathbf{m}$).*

Proof: see Appendix B.

Since $\mathbf{m}^*$ is unknown, [7] peels at *every* realizable iRM-set $\mathbf{m} \in \mathbb{M}$ ($\mathbb{M}$ the set of realizable iRM-sets) and returns the best, a global $1/i$-approximation, at the cost of $\Theta(|\mathbb{M}|)$ peelings. Besides, iRM-sets can only be pruned using flow networks. As such, pure peeling-based approximation is even slower than the exact algorithms [7]. A recent exact variant [16] further prunes the iRM-sets $\mathbf{m}$ with box-level bounds, but stays within the same paradigm: every surviving iRM-set is still solved by flow, so the number of subproblems remains data-dependent and each costs a flow computation.

The $i = 2$ advantage. At $i = 2$, directed/bipartite densest subgraph, the iRM-set collapses to a single scalar *ratio* ($m_2 = 1/m_1$, $m_1 \in [n^{-1/2}, n^{1/2}]$). A $(1{+}\varepsilon)$ approximation of the ratio suffices, so geometric spacing spans the whole interval with only $\mathcal{O}(\varepsilon^{-1} \log n)$ guesses, the "$\log n$ trick" [14] that replaces Charikar's $\Theta(n^2)$ ratio sweep [12], [13]. Each fixed-ratio subproblem is a supermodular program solved to $(1{-}\varepsilon)$ by accelerated coordinate descent [15], [17].

Algorithm 1: Cover-and-Conquer framework

Input: HIN G; meta-path $\mathcal{P}$
Output: A $\mathcal{P}$-family $\widehat{\mathcal{V}}$

```
Partition the iRM polytope Π_n into non-overlapping cells C_1, ..., C_K, V̂ ← ∅;
foreach cell C_k do
    pick a representative iRM-set m̂_k ∈ C_k;
    V_k ← SOLVE(G, P, m̂_k);
    if ρ(V_k) > ρ(V̂) then V̂ ← V_k;
return V̂;
```

Why this does not lift. The above optimisations presuppose *one-dimensionality* of the scalar ratio. The search exploits the *total order* of the ratio interval: a $(1{+}\varepsilon)$ change in the ratio moves the density only by $(1{+}\mathcal{O}(\varepsilon))$, so one geometric scan suffices and the order lets core decomposition and divide-and-conquer discard sub-intervals that cannot hold the optimum. Once $i - 1 > 1$, Property 1 makes the feasible iRM-sets a $(i-1)$-dimensional surface with *no* total order, and hence no axis to sweep and no sub-interval to prune. The optimum should therefore be *covered* in the hyperplane, not searched in a line. The subproblem solver fails for two reasons: at $i = 2$ the numerator is the *edge* count, a supermodular function the existing solvers exploit, whereas at $i > 2$ it is the *meta-path instance* count $|\mathcal{F}(S)|$, whose supermodularity is not a priori clear; and even granting it, the per-type iRM weights turn the denominator into a general i-type weighted size $\sum_j m_j |S_j|$, beyond the two-sided $m_1|S_1| + \frac{1}{m_1}|S_2|$ those solvers handle.

IV. Solution Overview

Our approach builds on a simple *cover-and-conquer* principle that decouples the two difficulties of DPSS: searching the iRM configuration space and solving each fixed-weight subproblem. Rather than enumerating every iRM-set, we partition the iRM polytope Π_n (§V) into a small number of *non-overlapping* cells, solve a *single* representative iRM-set in each, and return the densest family found.

Algorithm 1 is parameterized by two components, each supplied by a later section: a *partition* of Π_n into disjoint cells $C_1 \sqcup \cdots \sqcup C_K$ (§V), and a *cell solver* $\textsc{Solve}(G, \mathcal{P}, \mathbf{m})$ returning a $\mathcal{P}$-family for a fixed iRM-set $\mathbf{m}$ (§VI). It does not search for $\mathbf{m}^*$; it solves one representative per cell, on the premise that the cell containing $\mathbf{m}^*$ already yields a near-optimal family. Three questions decide whether the premise holds.

(Q1) Does a small partition bound the density error? The representative $\widehat{\mathbf{m}}_k$ need not be induced by a family, so the partition must be both *few* and *faithful*: solving at $\widehat{\mathbf{m}}_k$ rather than at the optimum's own weights must cost little density. §V delivers both: an η-net covers Π_n with only $\mathcal{O}((\log n/\sqrt{\eta})^{i-1})$ cells (polylogarithmic in n, against the $\mathcal{O}(|\mathbb{M}|)$ enumeration), each within a tunable loss factor $1 + \eta$ of the optimum (§V).

(Q2) Can each cell be solved near-optimally? At a fixed iRM-set the subproblem is still a hard densest-subgraph problem. §VI supplies a near-optimal cell solver returning, for a tunable δ, a family within a *solver factor* $\beta = 1 - \delta$ of the best at $\mathbf{m}$, running without ever materializing the meta-path instances (§VI-D).

(Q3) Do the two approximation errors compose? Both fall on each cell and multiply, and the cell holding $\mathbf{m}^*$ ties them to the global optimum: its *cell optimum* $\rho^*(C_k) := \max\{\rho(\mathcal{V}) : \mathrm{iRM}(\mathcal{V}) \in C_k\}$ ($\mathrm{iRM}(\mathcal{V})$ the iRM-set $\mathcal{V}$ induces) equals ρ^* and the realizable $\mathbf{m}^*$ makes $\mathcal{V}^*$ a competitor there, so that cell's SOLVE certifies density $\geq \frac{\beta}{1+\eta}\rho^*$, and reporting the densest family over all cells improves it. The result is a $\frac{\beta}{1+\eta} = \frac{1-\delta}{1+\eta}$ approximation, near-optimal for every fixed i.

V. From Enumeration to Covering

The cover promised in §III comes down to one fact: approximation does not need the *exact* optimal iRM-set $\mathbf{m}^*$, only one *close* to it, so that the exponential search for $\mathbf{m}^*$ collapses to solving at a few fixed *representatives*.

Closeness in *weights* yields closeness in *density*. Recall from §III that fixing the weights replaces the coupling geometric mean in the density by the linear *weighted size* $w_{\mathbf{m}}(S) = \sum_j m_j|S_j|$ (Eq. (2)), equal to ig at a family's own iRM weights. Solving instead at a nearby representative $\mathbf{m}'$ scores the optimum $\mathcal{V}^*$ at the *wrong* weights: each m_j^* becomes m_j', so (using $|V_j^*| = g(\mathcal{V}^*)/m_j^*$) its weighted size grows from $w_{\mathbf{m}^*}(\mathcal{V}^*) = i\,g(\mathcal{V}^*)$ to $w_{\mathbf{m}'}(\mathcal{V}^*) = \sum_j m_j'|V_j^*|$. Since the solver reads density as the *weighted density* $i\,h_{\mathbf{m}}(S) = i|\mathcal{F}(S)|/w_{\mathbf{m}}(S)$ (exact at a family's own weights), the density it assigns $\mathcal{V}^*$ at $\mathbf{m}'$ is deflated by a *loss factor* ℓ:

$$\frac{\rho_{\mathbf{m}'}(\mathcal{V}^*)}{\rho^*} = \frac{w_{\mathbf{m}^*}(\mathcal{V}^*)}{w_{\mathbf{m}'}(\mathcal{V}^*)} = \frac{i\,g(\mathcal{V}^*)}{\sum_j m_j'|V_j^*|} = (\tfrac{1}{i}\sum_j m_j'/m_j^*)^{-1} = \frac{1}{\ell}.$$

Here $\ell := \frac{1}{i}\sum_j m_j'/m_j^* \geq 1$, with equality iff $\mathbf{m}' = \mathbf{m}^*$, by AM–GM. The optimum thus attains only ρ^*/ℓ at the representative; the closer $\mathbf{m}'$ to $\mathbf{m}^*$, the closer ℓ to 1, so a small set of representatives covers $\mathbf{m}^*$ within a controlled density loss. Since ℓ is governed by the per-axis ratios m_j'/m_j^*, covering *every* possible $\mathbf{m}^*$ at once means distributing the representatives *multiplicatively*, so that each feasible iRM-set lies within a small ratio of one on every axis.

The change of coordinates. A multiplicative grid is awkward in native coordinates. $\prod_j m_j = 1$ is a *curved* surface that no axis-aligned grid meets, and the closeness we need is the *multiplicative* ratio m_j'/m_j, so a uniformly spaced grid in $\mathbf{m}$ is needlessly fine near the origin and uselessly coarse far from it. Passing to *logarithms* $x_j = \ln m_j$ cures both: feasibility collapses to the *additive* $\sum_j x_j = 0$, a flat $(i-1)$-dimensional hyperplane, and a multiplicative ratio becomes an additive displacement.

Roadmap. §V-A bounds the targets in a flat $(i-1)$-polytope Π_n; §V-B covers it with an η-net at a coarse $\sqrt{\eta}$ pitch ($1+\eta$ loss); §V-C turns weight-closeness into a density $\frac{\beta}{1+\eta}$ guarantee; and §V-D shrinks the net to the sub-area where the optimum can occupy.

A. The Bounded Log-iRM Polytope

Write an iRM-set as $\mathbf{m} = (m_1, \ldots, m_i)$ and pass to logarithmic coordinates $x_j = \ln m_j$, $\mathbf{x} = (x_1, \ldots, x_i)$. By the feasibility condition of Property 1, $\prod_{j=1}^i m_j = 1$ linearizes to $\sum_{j=1}^i x_j = 0$.

Lemma 1. *In logarithmic coordinates the feasible iRM-sets form the $(i-1)$-dimensional affine subspace (a hyperplane)*

$$\mathcal{H} = \Big\{\mathbf{x} \in \mathbb{R}^i : \textstyle\sum_{j=1}^i x_j = 0\Big\}.$$

Proof. Under the bijection $\mathbf{x} \mapsto e^{\mathbf{x}}$, $\prod_j m_j = 1$ is equivalent to $\sum_j x_j = 0$, one linear constraint of codimension one. □

So the i coordinates carry only $i-1$ free parameters. For $\langle \mathsf{C}, \mathsf{G}, \mathsf{D}\rangle$ ($i = 3$), Figure 3(a) draws the resulting 2-dimensional plane, which is a finite *hexagon*. We settle its boundary below.

Each m_j is located between the smallest and largest values, and under logarithms, the bound is a hypercube. The lemma below makes this precise.

Lemma 2. *Every feasible iRM-set satisfies*[2]

$$n^{-(i-1)/i} \;\leq\; m_j \;\leq\; n^{(i-1)/i} \qquad \forall\, j \in [i].$$

Proof. $m_j = \big(\prod_{k\neq j}|V_k|/|V_j|^{i-1}\big)^{1/i}$; with $1 \leq |V_k| \leq n$ the numerator is $\leq n^{i-1}$ and the denominator ≥ 1, giving the upper bound, and symmetrically the lower bound. □

Taking logarithms, every feasible point lies in the hypercube

$$B_n = \Big[-\tfrac{i-1}{i}\ln n,\ \tfrac{i-1}{i}\ln n\Big]^i,$$

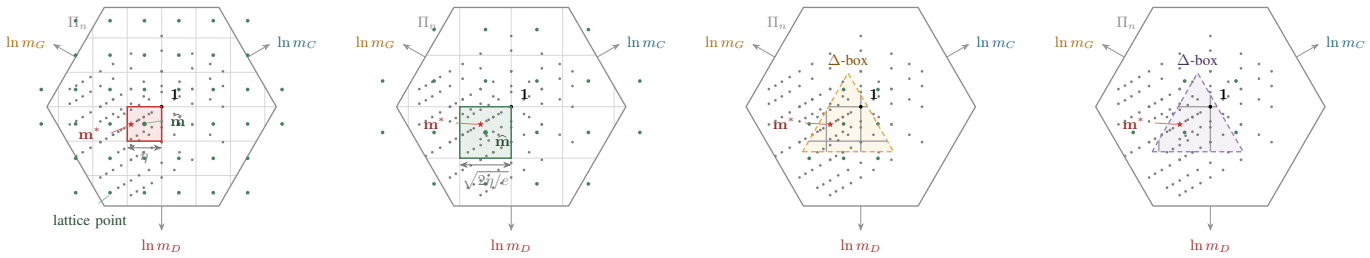
The bound treats each layer as free in $[1, n]$; the actual budget $\sum_k |V_k| \leq n$ forbids those extremal profiles (a coordinate reaches $n^{(i-1)/i}$ only if $i-1$ parts are each of size n), so B_n strictly over-encloses the feasible region, which only tightens the cover. Intersecting the hypercube with the hyperplane yields the feasible region $\Pi_n = B_n \cap \mathcal{H}$, a bounded $(i-1)$-dimensional polytope that contains every feasible iRM-set. That settles boundedness.

At $i = 3$, Π_n is the hypercube B_n sliced perpendicular to its main diagonal $(1, 1, 1)$, a regular hexagon; in general it is a $(i-1)$-polytope with up to $2i$ facets.

B. Covering with the η-Net

Now we are ready to lay the cover. Two constraints shape the grid: every grid point must be a *feasible* weight vector ($\prod_j m_j = 1$), and we must be able to *round* an arbitrary target onto it. A naive axis-aligned grid in the i raw log-coordinates meets neither: a generic point has $\sum_j x_j \neq 0$, off the hyperplane $\mathcal{H}$, so rounding to it leaves $\mathcal{H}$ and re-projecting

[2]The uniform cap $|V_k| \leq n$ is for readability; with per-type sizes $n_k = |V(A_k)|$ the same computation yields the per-axis bounds $n_j^{-(i-1)/i} \leq m_j \leq \big(\prod_{k\neq j} n_k\big)^{1/i}$.



(a) polytope Π_n, step $\ln(1+\eta)$ (b) the η-net, step $\sqrt{2\eta/e}$ (c) Δ-box + step η (d) Δ-box + step $\sqrt{2\eta/e}$

Fig. 3. The log-iRM polytope and covering nets

reintroduces error on every coordinate. Laying the grid *on* $\mathcal{H}$, as an integer lattice, fixes both at once.

Definition 1 (Zero-sum log-lattice)**.** *For a step $s > 0$ let*

$$\Lambda_s = \Big\{(k_1 s, \dots, k_i s) : k \in \mathbb{Z}^i,\ \textstyle\sum_j k_j = 0\Big\} \cap B_n^{+s},$$
$$B_n^{+s} = \big[-\tfrac{i-1}{i}\ln n - s,\ \tfrac{i-1}{i}\ln n + s\big]^i,$$

i.e. the lattice points whose log-coordinate $x_j = k_j s$ is an integer multiple of the step s, with the index $k_j = x_j/s \in \mathbb{Z}$ and $\sum_j k_j = 0$ keeping $\sum_j x_j = 0$ (the point on $\mathcal{H}$). The box is grown by one step per side, the smallest enlargement whose cells cover B_n, so every rounding image of a point in B_n is retained. Let $\mathcal{M}^s = \{(e^{x_1}, \dots, e^{x_i}) : \mathbf{x} \in \Lambda_s\}$ be its image in weight space. Every point of $\mathcal{M}^s$ satisfies $\prod_j m_j = e^{\sum_j x_j} = 1$ exactly, *with no projection step.*

Lemma 3 (Zero-sum rounding)**.** *For every feasible target $\mathbf{x}^* \in B_n$ with $\sum_j x_j^* = 0$ there is $\boldsymbol{\lambda} \in \Lambda_s$ with $|\lambda_j - x_j^*| \le s$ for all j.*

Proof. Round each coordinate, $k_j = \lfloor x_j^*/s + \frac{1}{2} \rfloor$, with error $e_j = k_j s - x_j^* \in [-s/2, s/2]$; the defect $r = \sum_j k_j \in \mathbb{Z}$ has $|r| \le i/2$. If $r \neq 0$, adjust $|r|$ coordinates by ∓ 1 (possible since at least $|r|$ errors share its sign), so each adjusted error lands in $[-s, s)$, the rest stay in $[-s/2, s/2]$, and now $\sum_j k_j = 0$. Every final error has magnitude at most s. □

Choosing the granularity. We ask of the lattice one thing: rounding any feasible target to its nearest net point should keep the density loss to a factor $1 + \eta$. A coarse pitch can meet it, as the next definition and theorem show.

Definition 2 (η-net)**.** *For $\eta \in (0, 1]$, set $s(\eta) = \sqrt{2\eta/e}$ and let $\mathcal{M}_\eta = \mathcal{M}^{s(\eta)}$ (the* second-order net*) be the zero-sum log-lattice of Definition 1 at this step.*

Theorem 2 (A small η-net)**.** *For $\eta \in (0, 1]$, the net $\mathcal{M}_\eta$ has size $\mathcal{O}\big((\log n/\sqrt{\eta})^{i-1}\big)$, and for every feasible $\mathbf{m}^*$ it contains a representative $\mathbf{m}'$ with $\prod_j m'_j = 1$ whose loss factor $\ell = \frac{1}{i}\sum_j e^{\theta_j}$, $\theta_j = \ln(m'_j/m^*_j)$, equals $1 + \Theta(\|\boldsymbol{\theta}\|^2)$ and satisfies $\ell \le 1 + \eta$.*

Proof. $\Lambda_{s(\eta)}$ is $(i-1)$-dimensional (the constraint $\sum_j k_j = 0$ removes one degree of freedom) and each free coordinate ranges over $\mathcal{O}(\log n/s(\eta)) = \mathcal{O}(\log n/\sqrt{\eta})$ values (Lemma 2), giving the size. Feasibility is Definition 1; the closeness $\|\boldsymbol{\theta}\|_\infty \le s(\eta)$ is Lemma 3 at $\mathbf{x}^* = \ln \mathbf{m}^*$. For the loss, the balance $\sum_j \theta_j = 0$ cancels the linear term,

$$\ell = \tfrac{1}{i}\textstyle\sum_j e^{\theta_j} = 1 + \underbrace{\tfrac{1}{i}\sum_j \theta_j}_{=0} + \tfrac{1}{2i}\textstyle\sum_j \theta_j^2 + \cdots = 1 + \Theta\big(\|\boldsymbol{\theta}\|^2\big),$$

the curvature of the AM–GM gap at equality. For the bound, the elementary inequality $e^t \le 1 + t + \frac{t^2}{2}e^{|t|}$ (valid for every real t), applied at $t = \theta_j$ and averaged over j, gives $\ell \le 1 + \frac{1}{i}\sum_j \theta_j + \frac{1}{2i}\sum_j \theta_j^2 e^{|\theta_j|}$; the linear sum vanishes by $\sum_j \theta_j = 0$, and with $|\theta_j| \le s(\eta) \le 1$ each $e^{|\theta_j|} \le e$, so $\ell \le 1 + \frac{e}{2}s(\eta)^2 = 1 + \eta$. □

An intuitive baseline. The same $1 + \eta$ loss is met more straightforwardly by pinning *each* ratio m'_j/m^*_j within $1 + \eta$ on its own, the *first-order net* at pitch $s = \ln(1 + \eta) = \Theta(\eta)$. Being quadratically finer, it lays $\eta^{-(i-1)/2}$ times as many cells (Figure 3(a) vs. (b)); we keep it only as a baseline.

These net points are the representatives $\widehat{\mathbf{m}}_k$ that Algorithm 1 solves. Each is the *center* of its cell, the lattice point every iRM-set there rounds to (Lemma 3), so one SOLVE covers the whole cell within $1 + \eta$. Since $\mathbf{m}^*$ is unknown, the algorithm solves at *every* net point and returns the densest family; the cell covering $\mathbf{m}^*$ is guaranteed among them, so the global best inherits the $1 + \eta$ bound.

On the running instance of Figure 2, the 110 distinct feasible iRM-sets are covered by 40 net cells for the baseline η net, centered at green lattice points (Figure 3(a)). There are only 20 net cells (Figure 3(b)) for the $\sqrt{2\eta/e}$ net. One SOLVE per cell instead of 110, secures a density loss of $1 + \eta$.

C. How Net Approximation Affects Subproblem Quality

Theorem 2 puts a representative $\mathbf{m}'$ within loss factor $\ell \le 1 + \eta$ in the cell of the unknown optimum $\mathbf{m}^*$. Running an *existing* fixed-weight solver, i.e., the peeling for the $1/i$ approximation algorithm, at $\mathbf{m}'$ turns this closeness in *weights* into closeness in *density*: the *approximate* weights the net hands to the solver cost only the tunable factor $1 + \eta$.

Lemma 4 (Peeling is ratio-robust)**.** *Let $\mathcal{V}^*$ be optimal with iRM-set $\mathbf{m}^*$, and $\mathbf{m}'$ feasible with loss factor $\frac{1}{i}\sum_j e^{\theta_j} \le 1 + \eta$, $\theta_j = \ln(m'_j/m^*_j)$ (Theorem 2). Peeling with $\mathbf{m}'$ on any subgraph containing $G(\mathcal{V}^*)$ yields $\widehat{\gamma}$, the largest density seen during the peel, with $\rho^* \le i(1 + \eta)\,\widehat{\gamma}$.*

Proof: see Appendix B.

Theorem 3 (Peeling over the net)**.** *Running the peeling once at every net point of $\mathcal{M}_\eta$ and returning the densest $\mathcal{P}$-family is a $\frac{1}{i(1+\eta)}$-approximation to ρ^*, with $\mathcal{O}((\log n/\sqrt{\eta})^{i-1})$ peeling runs.*

The correctness is clear: it is the solver-agnostic reduction above at $\beta = \frac{1}{i}$. Lemma 4 gives the per-cell guarantee $\widehat{\gamma} \geq \rho^*/(i(1+\eta))$ in $\mathbf{m}^*$'s cell. Theorem 2 bounds the number of peeling runs.

D. Localizing the Hypercube

The net size carries a factor $(\log n)^{i-1}$ from the worst-case hypercube width $B_n = \mathcal{O}(\log n)$ per layer (Lemma 2). The *optimum's* weights, though, cannot roam the whole hypercube: an iRM-weight is not free, and at the optimum the j-th weight is pinned to how instance-rich a typical type-j vertex is. Precisely, m_j^* is the *average* number of meta-path instances through a type-j vertex of $\mathcal{V}^*$, divided by ρ^*. Writing $\Delta_j = \max_{v \in V(A_j)} P(v, G)$ for the *largest* such per-vertex count, an average never exceeds its maximum, so Δ_j caps how far the optimum can lean on layer j.

Lemma 5 (Δ-box)**.** *Assume $\rho^* > 0$. For the optimal family $\mathcal{V}^*$, $m_j^* = \bar{P}_j/\rho^*$ where $\bar{P}_j = |\mathcal{F}(\mathcal{V}^*)|/|V_j^*|$, and $1 \leq \bar{P}_j \leq \Delta_j$. Hence $m_j^* \in [1/\rho^*,\ \Delta_j/\rho^*]$, an interval of log-width $\ln \Delta_j$.*

Proof. $\bar{P}_j/m_j^* = |\mathcal{F}(\mathcal{V}^*)|/g(\mathcal{V}^*) = \rho^*$ gives the identity. Since $\sum_{v \in V_j^*} P(v, G(\mathcal{V}^*)) = |\mathcal{F}(\mathcal{V}^*)|$, the average $\bar{P}_j \leq \Delta_j$; and $\bar{P}_j \geq 1$, as a type-j vertex in no instance could be deleted to raise the density (or, if alone, forces $\rho^* = 0$). □

So the optimum's j-th weight is confined to a band of log-width only $\ln \Delta_j$, *independent of* n. Its location depends on the unknown ρ^*, but feasibility brackets it regardless. Taking the product of $m_j^* = \bar{P}_j/\rho^*$ over j and using $\prod_j m_j^* = 1$ gives $\rho^* = \left(\prod_j \bar{P}_j\right)^{1/i}$, the geometric mean of the $\bar{P}_j$; since each $\bar{P}_j \in [1, \Delta_j] \subseteq [1, \Delta]$ ($\Delta = \max_j \Delta_j$) and a geometric mean lies between its smallest and largest terms, $\rho^* \in [1, \Delta]$. This bracket alone fixes the box, with no estimate of ρ^*: in $m_j^* = \bar{P}_j/\rho^*$ the numerator is at most Δ_j and at least 1, so $\rho^* \geq 1$ gives $m_j^* \leq \Delta_j$ while $\rho^* \leq \Delta$ gives $m_j^* \geq 1/\Delta$, hence $\ln m_j^* \in [-\ln \Delta,\ \ln \Delta_j]$ *whatever* ρ^* is. The band thus *provably* contains $\mathbf{m}^*$: only the bracket $\rho^* \in [1, \Delta]$ enters, not an estimate. This bracket only tightens during the search, where the best density found so far, $\hat{\rho} \leq \rho^*$, sharpens it to $\rho^* \in [\hat{\rho}, \Delta]$ and shrinks the box to $m_j^* \leq \Delta_j/\hat{\rho}$ while still containing $\mathbf{m}^*$. Tiling this band in place of B_n (grown by one step per side, as in Definition 1) reduces the net size to $\mathcal{O}\big((\min\{\log n, \log \Delta\}/\sqrt{\eta})^{i-1}\big)$.

Δ plus net. The net and the Δ-box touch *different* parts of the per-axis base and so compose without interference. The net (§V-B) sets the *pitch* $\mathcal{O}(\sqrt{\eta})$, the Δ-box the *range* $\min\{\log n, \log \Delta\}$. Together they are the cover Algorithm 1 sweeps, one fixed-weight SOLVE per cell, each within $1 + \eta$ of the optimum's (§IV), and the net we carry into §VI.

On the running example. Figure 3(c) draws the Δ-box: the slab $m_j \in [1/\rho^*, \Delta_j/\rho^*]$ (here $(\Delta_1, \Delta_2, \Delta_3) = (3, 4, 5)$) meets the product-one plane in a small *triangle*, with $\mathbf{m}^*$ inside. Overlaying the coarse $\sqrt{2\eta/e}$ pitch on it (Figure 3(d)) leaves only 5 cells, a tiny fraction of the 110 feasible iRM-sets naive enumeration would visit, yet the cell of $\mathbf{m}^*$ is among them.

VI. Near-Optimal Approximation for $i > 2$

What remains for a near-optimal algorithm is to push the $\frac{1}{i}$ factor to $1 - \delta$. The road has three steps: the fixed-weight subproblem is a weighted supermodular densest-subgraph problem in the auxiliary objective $h_{\mathbf{m}}$ (§VI-A); a native solver makes it near-optimal *in* $h_{\mathbf{m}}$ (§VI-B); and an AM–GM bridge, tight at the optimum's own weight, carries that near-optimality *to the density* ρ (§VI-C).

Why the near-optimal solver stopped at $i = 2$. The fixed-weight subproblem is a weighted supermodular densest-subgraph problem (Theorem 4, below), so one might expect the near-optimal supermodular solvers [18]–[20] to finish the job; the obstacle is the *per-type weights*. Those solvers target the *cardinality* ratio $f/|S|$, and the $i = 2$ methods target only the two-sided $m_1|S_1| + \frac{1}{m_1}|S_2|$ [14], [15], [17]; neither handles a general i-type denominator. The possible workaround, *materializing* $m_{\phi'(v)}$ integer copies of each v to feed an standard *unit-weight* peeler, also fails: the weights are real-valued and spread up to n^2, so exact copies do not exist and rounded ones inflate the graph and corrupt the peeling order.

A. Supermodular Densest Subgraph

Recall the fixed-weight subproblem from §III: for an iRM-set $\mathbf{m}$, maximize the auxiliary objective $h_{\mathbf{m}}(S) = f(S)/w_{\mathbf{m}}(S)$ over $S \subseteq U$, with $f(S) = |\mathcal{F}(S)|$ and the modular weighted size $w_{\mathbf{m}}(S) = \sum_j m_j |S_j|$ (the weighted density of §V-C is $i\, h_{\mathbf{m}}$). The denominator $w_{\mathbf{m}}$ is already modular; the numerator is supermodular.

Lemma 6. *$f : 2^U \to \mathbb{Z}_{\geq 0}$, $f(S) = |\mathcal{F}(S)|$, is normalized ($f(\emptyset) = 0$), monotone, and* supermodular.

Proof. Normalization and monotonicity are immediate. For supermodularity, fix $S \subseteq S'$, $v \notin S'$: every instance in the marginal $f(S \cup \{v\}) - f(S)$ passes through v with its other vertices in $S \subseteq S'$, so it is also in $f(S' \cup \{v\}) - f(S')$. □

A supermodular numerator over a modular denominator is thus the problem below.

Theorem 4. *For every feasible iRM-set $\mathbf{m}$, maximizing $h_{\mathbf{m}}(S)$ over $S \subseteq U$ is an instance of weighted supermodular densest subgraph (immediate from Lemma 6 and the modularity of $w_{\mathbf{m}}$).*

B. A Near-Optimal Weighted Solver

As discussed, the cardinality-ratio solvers do not apply, and the workaround fails, so we need a native solver. We adapt SUPER-GREEDY++ [19] (the supermodular form of Greedy++ [18]), which peels the vertex of least load-augmented degree $\sigma(v) + \deg_S(v)$ and charges its degree back across passes, in two changes as follows. *(i) Weight the criterion:* delete the vertex of least *weighted* incidence $P(v, \cdot)/m_{\phi'(v)}$, putting the per-type weight into the deletion order. *(ii) Charge and repeat:* carry a per-vertex load across passes, adding a

vertex's incidence when it is deleted, so a vertex peeled too early is spared next pass and the order self-corrects.

Why repeat: a rising floor meets a descending ceiling. Any family peeling returns a *floor*, an achievable density that prior peeling (Theorem 1) keeps within a fixed factor of optimal, though a single pass gives no per-instance certificate. Repeating adds a *ceiling*: the averaged load, peaked against the per-type weights, upper-bounds the optimum. One deletion order does both, sparing instance-rich vertices to lift the floor while averaging the charges to lower the ceiling; once they meet, the family in hand is certified near-optimal. The floor is inherited from the previous peeling, while the ceiling and its convergence are formalized in Proposition 1.

Proposition 1 (Iterative Peeling)**.** *Peeling by the load-augmented key* $\big(\sigma(v) + P(v, G(S))\big)/m_{\phi'(v)}$ *with* $\sigma(v)$ += $P(v, G(S))$ *charged at each deletion (ideas (i)–(ii)), reporting the densest $\mathcal{P}$-family over all prefixes of all passes, returns S with $h_{\mathbf{m}}(S) \geq (1-\delta)\, h^*_{\mathbf{m}}$ ($h^*_{\mathbf{m}} = \max_{\emptyset \neq S' \subseteq U} h_{\mathbf{m}}(S')$) in $T = \mathcal{O}\big(\Delta_{\mathbf{m}} \ln |U| \,/\, (\delta^2\, h^*_{\mathbf{m}})\big)$ passes, where $\Delta_{\mathbf{m}} = \max_{v \in U} P(v, G)/m_{\phi'(v)} = \max_j \Delta_j/m_j$ is the* weighted width.

Proof sketch. The argument rests on three lemmas, proved in full in Appendix A; we state what each supplies and how they combine.

(i) Base membership. No matter which key chooses the vertex to delete, a pass charges each deleted vertex its marginal $P(v, G(S)) = f(v \,|\, S \setminus v) := f(S) - f(S \setminus v)$, so read in reverse deletion order, the charges form a *greedy base* of the supermodular f (greedy bases and exact linear minimization). Hence each pass's charge lies in the base polytope $B(f) = \{x \geq 0 : x(S) \geq f(S)\ \forall S,\ x(U) = f(U)\}$ (writing x_v for the coordinates of $x \in \mathbb{R}^U$ and $x(S) = \sum_{v \in S} x_v$), and so does the averaged load $\bar{x}_t = \sigma_t/t$, σ_t the total charge after t passes, by convexity (per-pass base membership).

(ii) A computable certificate. Weak duality on $B(f)$ converts any averaged load into a bound: for $x \in B(f)$ and $\emptyset \neq S$, $f(S) \leq x(S)$ gives $h_{\mathbf{m}}(S) \leq \max_v x_v/m_{\phi'(v)}$, so the gap $\hat{g}_t = \max_v \bar{x}_t(v)/m_{\phi'(v)} - \max_{\text{prefix } S} h_{\mathbf{m}}(S)$ is nonnegative and computable, and $\hat{g}_t \leq \delta\, h^*_{\mathbf{m}}$ certifies the reported family $(1-\delta)$-dense *unconditionally*.

(iii) The gap closes at the stated rate. The anchor is the weighted quasi-min-max $h^*_{\mathbf{m}} = \min_{x \in B(f)} \max_v x_v/m_{\phi'(v)}$, the per-type-weighted counterpart of the cardinality min–max, obtained via the Lovász extension of f and von Neumann's minimax. Against this anchor, the convergence analysis of SUPER-GREEDY++ [19] carries over, once loads, marginals and key are read *per unit weight*: the weights enter only through $h^*_{\mathbf{m}}$ and the width $\Delta_{\mathbf{m}}$, giving $\min_{t \leq T} \hat{g}_t \leq \delta\, h^*_{\mathbf{m}}$ at the stated T. On the Δ-box-localized net the width-to-value ratio collapses to our own parameters, so the uniform budget $T = \mathcal{O}(i\, \Delta \ln |U|/\delta^2)$ suffices ($1 + \eta \leq 2$ is absorbed into the constant), the $1/\delta^2$ exponent being proper to peeling-type schemes (Remark A.3). At T=1, $\sigma \equiv 0$ and the key is $P(v, G(S))/m_{\phi'(v)}$, verbatim the reweighted peel of Theorem 1; the load carried across passes is the lift to $(1-\delta)$. □

By now, we settle the subproblem in the *weighted objective*: for any fixed weight $\mathbf{m}$, the solver returns a family within $(1-\delta)$ of $h^*_{\mathbf{m}}$. Our target, however, is the density ρ, not $h_{\mathbf{m}}$, which will be addressed in §VI-C.

C. From $h_{\mathbf{m}}$ to the Density ρ

We now convert near-optimality in $h_{\mathbf{m}}$ into near-optimality in the density ρ.

Lemma 7. *For every $S \subseteq U$ with each $S_j \neq \emptyset$, $\rho(S) \geq i\, h_{\mathbf{m}}(S)$, with equality iff $\mathbf{m}$ is S's own iRM-set.*

Proof. By the AM–GM bound behind the loss factor of §V, $w_{\mathbf{m}}(S) = \sum_j m_j |S_j| \geq i(\prod_j |S_j|)^{1/i} = i\, g(S)$ (using $\prod_j m_j = 1$); dividing into $f(S)$ gives $h_{\mathbf{m}}(S) \leq \rho(S)/i$. The bound is tight iff the $m_j|S_j|$ are all equal, i.e. iff $\mathbf{m}$ is S's own iRM-set. □

At the optimum's own iRM-set $\mathbf{m}^*$ the factor i thus cancels, $i\, h_{\mathbf{m}^*}(\mathcal{V}^*) = \rho^*$, so solving there would return $\rho(S') \geq (1-\delta)\rho^*$ directly. Since $\mathbf{m}^*$ is unknown, the net supplies only a representative $\mathbf{m}'$, at the tunable cost $1 + \eta$:

Lemma 8. *Let $\mathbf{m}'$ be the net representative of $\mathbf{m}^*$'s cell, with displacements $\theta_j = \ln(m'_j/m^*_j)$ and $\frac{1}{i}\sum_j e^{\theta_j} \leq 1 + \eta$ (Theorem 2). Then $i\, h_{\mathbf{m}'}(\mathcal{V}^*) \geq \rho^*/(1+\eta)$.*

Proof. This is the loss-factor identity of §V: scoring $\mathcal{V}^*$ at $\mathbf{m}'$ deflates its density by exactly $\ell = \frac{1}{i}\sum_j e^{\theta_j}$, i.e. $i\, h_{\mathbf{m}'}(\mathcal{V}^*) = \rho^*/\ell$, and $\ell \leq 1 + \eta$ by Theorem 2. □

Theorem 5. *Running the solver of Proposition 1 once at every net point of the net $\mathcal{M}_\eta$ (Definition 2), localized by the Δ-box (Lemma 5), and returning the densest $\mathcal{P}$-family $\widehat{\mathcal{V}}$ yields*

$$\rho(\widehat{\mathcal{V}}) \;\geq\; \frac{1-\delta}{1+\eta}\,\rho^*,$$

using $\mathcal{O}\big((\min\{\log n, \log \Delta\}/\sqrt{\eta})^{i-1}\big)$ subproblem solvers, where $\Delta = \max_j \Delta_j$.

Proof. The cell containing $\mathbf{m}^*$ has a representative $\mathbf{m}' \in \mathcal{M}_\eta$ with $\prod_j m'_j = 1$ and loss factor $\leq 1 + \eta$ (Theorem 2); by Lemma 5 the localized box contains $\mathbf{m}^*$, hence (being tiled with one extra step per side, Definition 1) that cell. Let S' be the solver's output at $\mathbf{m}'$. Lemma 7's hypothesis holds at S': by Lemma 8 $i\, h^*_{\mathbf{m}'} \geq i\, h_{\mathbf{m}'}(\mathcal{V}^*) \geq \rho^*/(1+\eta) > 0$, so $h_{\mathbf{m}'}(S') \geq (1-\delta) h^*_{\mathbf{m}'} > 0$, hence $f(S') > 0$, and since an instance occupies every position, every layer of S' is nonempty. Chaining these,

$$\rho(S') \overset{\text{Lem. 7}}{\geq} i\, h_{\mathbf{m}'}(S') \overset{(1-\delta)}{\geq} (1-\delta)\, i \max_S h_{\mathbf{m}'}(S)$$
$$\geq\; (1-\delta)\, i\, h_{\mathbf{m}'}(\mathcal{V}^*) \overset{\text{Lem. 8}}{\geq} \frac{1-\delta}{1+\eta}\,\rho^*.$$

Reporting the densest family over all net points only improves the bound, and the number of solvers is that of Theorem 2 over the localized box. □

Corollary 1 (PTAS for fixed i)**.** *For every fixed i and $\zeta \in (0,1)$, taking $\delta = \eta = \zeta/3$ gives a $(1-\zeta)$-approximation in time polynomial in $|G|$, a PTAS, not an FPTAS.*

Proof: see Appendix B.

On the running example. The optimum is the $(4, 2, 2)$ module $\mathcal{V}^*$ of Example 2 ($\rho^* \approx 2.78$), interleaved with two decoy fans g_1, g_3 of loosely attached paths. Run the low-cost $\frac{1}{i}$ peel (§V-C) at *every* net weight: none isolates the module in a single pass, so the best family it sweeps out is the whole graph (≈ 2.58). The supermodular solver with $\delta = 0.01$ of this section, maximizing $h_{\mathbf{m}}$ to $(1 - \delta)$ at the module's own weight, instead returns $\mathcal{V}^*$ exactly. This 2.78-against-2.58 gap *persists under every net refinement*, a clear gap.

D. Running the Solver Instance-Free

The previous sections cut the *number* of subproblems; a second bottleneck hides inside each. Every peeling solver reads the instance set $\mathcal{F}(\mathcal{V})$, which for a meta-path can far exceed the graph ($|\mathcal{F}(\mathcal{V})| = \mathcal{O}(\prod_j |V_j|)$ against $|E|$ edges), and materializing it would be wasteful, which we show is unnecessary.

The factorization. Fix $\mathcal{P} = \langle A_1, \ldots, A_i \rangle$ and $\mathcal{V}$. For a vertex $v \in V(A_j)$ let $L(v)$ be the number of partial instances of the prefix $(A_1, \ldots, A_j)$ that *end* at v, and $R(v)$ the number of partial instances of the suffix $(A_j, \ldots, A_i)$ that *start* at v, the ways to reach v from the left, and to continue from v to the right.

Observation 1. $\forall v$ of $G(\mathcal{V})$, $P(v, G(\mathcal{V})) = L(v)\, R(v)$.

Maintaining counts. The two count families are themselves simple path dynamic programs. Writing $N^-(v)$ and $N^+(v)$ for the neighbors of v in the preceding and following layers,

$$L(v) = \sum_{u \in N^-(v)} L(u) \;\; (j > 1), \quad R(v) = \sum_{u \in N^+(v)} R(u) \;\; (j < i),$$

with $L(v) = 1$ on the first layer and $R(v) = 1$ on the last.

Observation 2. All counts $\{L(v), R(v)\}_v$ can be computed by one forward and one backward sweep over $G(\mathcal{V})$, in $\mathcal{O}(|E(G(\mathcal{V}))|)$ time and space: by the recurrences above, the forward sweep fills L in increasing j and the backward sweep fills R in decreasing j, each reading every edge once.

Instance-free peeling. ITERATIVE PEELING reads the instances only through its key $(\sigma(v) + P(v, G(S)))/m_{\phi'(v)}$, and both parts run on $G(\mathcal{V})$ alone: the loads $\sigma(v)$ are $\mathcal{O}(|U|)$ scalars updated in place, and the incidence factors as $P(v, G(S)) = L(v)R(v)$ (Observation 1), maintained by the two sweeps of Observation 2 and refreshed on deletion by propagating along incident edges. A pass thus keeps the keys $\big(\sigma(v) + L(v)R(v)\big)/m_{\phi'(v)}$ in a heap and repeatedly extracts the minimum, deletes, and propagates, the *same* execution as the enumerated solver (since $L(v)R(v) = P(v, G(S))$), so its $(1 - \delta)$ guarantee carries over with total state $\mathcal{O}(|E|)$, independent of $|\mathcal{F}(\mathcal{V})|$.

End-to-end cost. Two quantities, scaling differently in n, govern the cost. The *number* of subproblems is the cell count $K = \mathcal{O}\big((\min\{\log n, \log \Delta\}/\sqrt{\eta})^{i-1}\big)$, polylogarithmic in n and Δ-adaptive (Lemma 5), saving over the $\mathcal{O}(|\mathbb{M}|)$ enumeration of prior work. The *cost per subproblem* is polynomial: on the Δ-box-localized net a cell runs $T = \mathcal{O}(i\, \Delta \ln |U|/\delta^2)$ passes (Proposition 1), each costing $\mathcal{O}(|E|\, \Delta_{\text{prop}} + |U| \log |U|)$ with $|E| = |E(G(\mathcal{V}))|$, $|U| = \sum_j |V(A_j)|$, and $\Delta_{\text{prop}} \le |E|$ the maximum number of times one pass's count propagation re-reads a single edge. The search therefore runs in $\mathcal{O}\big(K\, T\, (|E|\, \Delta_{\text{prop}} + |U| \log |U|)\big)$.

Discussion. The bound on T is worst-case. For the simple densest subgraph, Greedy++'s convergence was conjectured by [18] and later proved [19]; faster first-order and coordinate-descent methods [17], [20] carry per-edge state that forfeits instance-freeness. In practice each solver stops far sooner, and the ceiling carries across representatives $\mathbf{m}$ (its averaged load is weight-free), trimming the search further (§VIII); the details are given in Appendix A (Remark A.1).

VII. VERTEX AND CELL PRUNING

Building on the supermodular reformulation of §VI, we propose a *per-type* incidence-*floor* vertex pruning that shrinks the graph *before* any solver. Because the floor scales with each vertex's type weight $m_{\phi'(v)}$, it is sharper than the single, uniform density threshold of prior HIN vertex pruning [7].

Fix a feasible iRM-set $\mathbf{m}$ with the auxiliary objective $h_{\mathbf{m}} = f/w_{\mathbf{m}}$ over the ground set U of §VI, and write $h^*_{\mathbf{m}} = \max_{\emptyset \neq S \subseteq U} h_{\mathbf{m}}(S) > 0$ for its fixed-weight optimum. We show the lemma below.

Lemma 9 (Vertex pruning)**.** *With the notation above:* (i) Incidence floor. *Every maximizer S° and every $v \in S^\circ$ satisfy $P(v, G(S^\circ)) \ge h^*_{\mathbf{m}}\, m_{\phi'(v)}$, and hence $P(v, G) \ge h^*_{\mathbf{m}}\, m_{\phi'(v)}$.* (ii) Optimum-preserving peel. *For any $\lambda \le h^*_{\mathbf{m}}$, deleting every vertex with $P(v, G) < \lambda\, m_{\phi'(v)}$ removes no vertex of any $h_{\mathbf{m}}$-optimal S°; iterating the deletion on the shrinking graph against the same λ likewise removes none, and leaves $h^*_{\mathbf{m}}$ unchanged.*

Proof: see Appendix B.

The floor of Lemma 9 involves the unknown $h^*_{\mathbf{m}}$, but part (ii) needs only a *lower* bound on it: any $\lambda \le h^*_{\mathbf{m}}$ is a lower bound, and deleting $\{v : P(v, G) < \lambda\, m_{\phi'(v)}\}$ removes no vertex of an optimal family. The running search supplies one, its best auxiliary value $h_{\mathbf{m}}(S) \le h^*_{\mathbf{m}}$ so far, which only improves across the run [7]; the pruning uses this λ in place of $h^*_{\mathbf{m}}$. When the pruning empties a cell's graph, that cell is dropped, and the bound still holds.

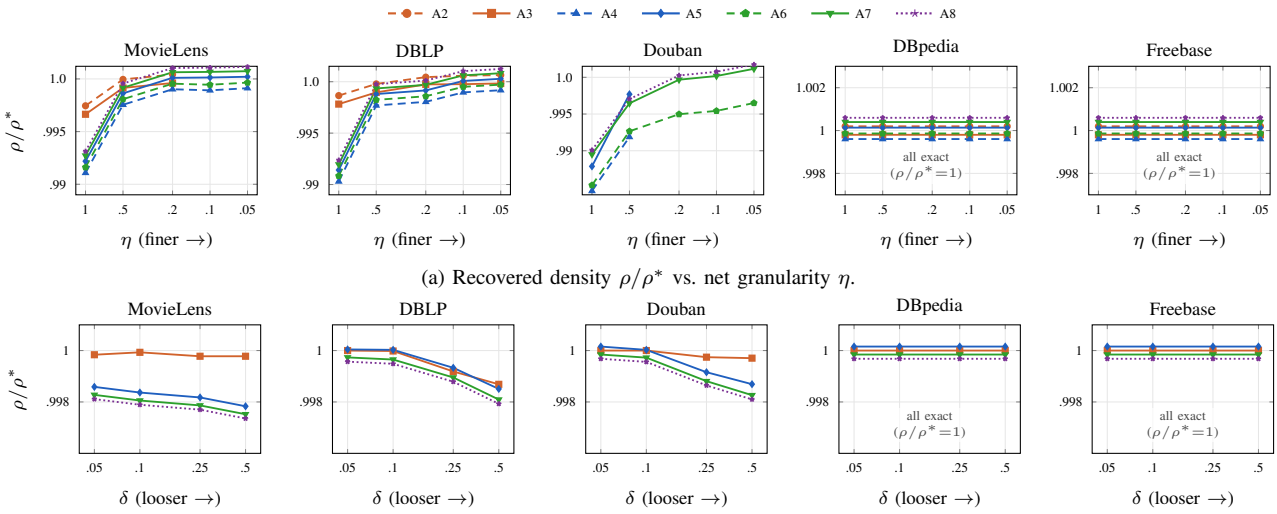
VIII. EXPERIMENTS

We now evaluate the effectiveness and efficiency of the proposed techniques.

A. Experimental Setup

Datasets. We use five benchmark HINs (Table II); HETIONET serves the case study (§VIII-B). Three are dense *core* benchmarks: MOVIELENS, DBLP, and DOUBAN [7]. Two are large *breadth* graphs rebuilt from public dumps, so their sizes differ from [7]: DBPEDIA (18.7M edges) and FREEBASE (433M edges), used to test scale and the limits of the cover on long, sparse meta-paths.

Queries (meta-paths). As in [7], from each schema we materialize a pool of meta-path subgraphs grouped by length i: every available $\mathcal{P}$ of $i \in \{3, 4\}$ on the core datasets, and a



(a) Recovered density ρ/ρ^* vs. net granularity η.

(b) Recovered density ρ/ρ^* vs. per-subproblem tolerance δ.

Fig. 4. Effectiveness evaluation.

TABLE II
DATASETS

Dataset	$\lvert\mathcal{A}\rvert$	$\lvert\mathcal{R}\rvert$	$\lvert V\rvert$	$\lvert E\rvert$	i
MovieLens	5	6	2.7K	0.10M	3–4
DBLP	4	3	37.8K	0.17M	3–4
Douban	6	6	37.6K	1.71M	3–4
DBpedia	412	661	6.31M	18.7M	3–9
Freebase	10.1K	14.8K	120M	433M	3–9

TABLE III
TESTED ALGORITHMS

	mark	Net	Per-cell solver	Guarantee
A1	—	exact (none)	exact	1
A2	-●-	first-order	$1/i$ peel	$\frac{1}{i(1+\eta)}$
A3	-■-	first-order	near-optimal	$\frac{1-\delta}{1+\eta}$ (conj.)
A4	-▲-	second-order	$1/i$ peel	$\frac{1}{i(1+\eta)}$
A5	-◆-	second-order	near-optimal	$\frac{1-\delta}{1+\eta}$ (conj.)
A6	-⬟-	local + second	$1/i$ peel	$\frac{1}{i(1+\eta)}$
A7	-▼-	local + second	near-optimal	$\frac{1-\delta}{1+\eta}$ (conj.)
A8	···✳···	local + second	near-optimal	$\frac{1-\delta}{1+\eta}$ (cert.)

sampled $i\in\{3,\ldots,9\}$ pool on the breadth datasets (140 each). Sampling is at the path-instance level, so each $\mathcal{P}$-partite graph is connected and has $\mathcal{P}$-instances. Compared to [7] and [16], our queries allow duplicated vertex types, which are often tougher.

Methods. We compare 8 algorithms, listed in Table III. A1, the state-of-the-art exact algorithm [16], gives the exact optimum ρ^* by enumerating necessary feasible iRM-sets with pruning (the approximation of [7] is consistently slower, checking every iRM-set[3]). A2–A8 factor the cover into a *net* (which weight sets are represented) and a per-cell *solver*: *first-order* = the $\ln(1+\eta)$-pitch baseline net (§V-B), *second-order* = the $\sqrt{2\eta/e}$-pitch net (§V-B), *local* = the Δ-box localization (§V-D), *near-optimal* = the supermodular solver (§VI). The near-optimal solver runs in two modes: a *conjectured* stop ("conj.": A3, A5, A7), a few passes reaching the near-optimal density in practice without a per-run guarantee [18], and a *certified* stop ("cert.": A8), run to the certificate of Proposition 1 that proves $\frac{1-\delta}{1+\eta}$ on every cell [19]; the proof of Proposition 1 discusses the two. A2–A8 all apply the pruning of §VII by default. Among A2-A8, A6 has the lowest time complexity, while A8 provides the guarantee.

[3]Therefore, we directly use the exact method as the baseline.

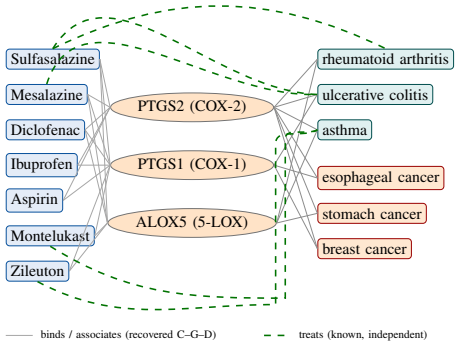


Fig. 5. Case study

Implementation and measures. All algorithms are implemented in C++ and compiled with GCC 13.3 (`-O2`) on an Apple M5 Max machine. We measure the algorithm runtime as the total CPU time, excluding the I/O cost of loading the graph into main memory; a cut-off time of 800 s and a 48 GB memory cap are set, and a non-completion is denoted ∞. Each reported number is averaged over the dataset's query pool. Code and data are available at https://github.com/S1mpleCod/Near-optimal-approximation-for-DPPS.

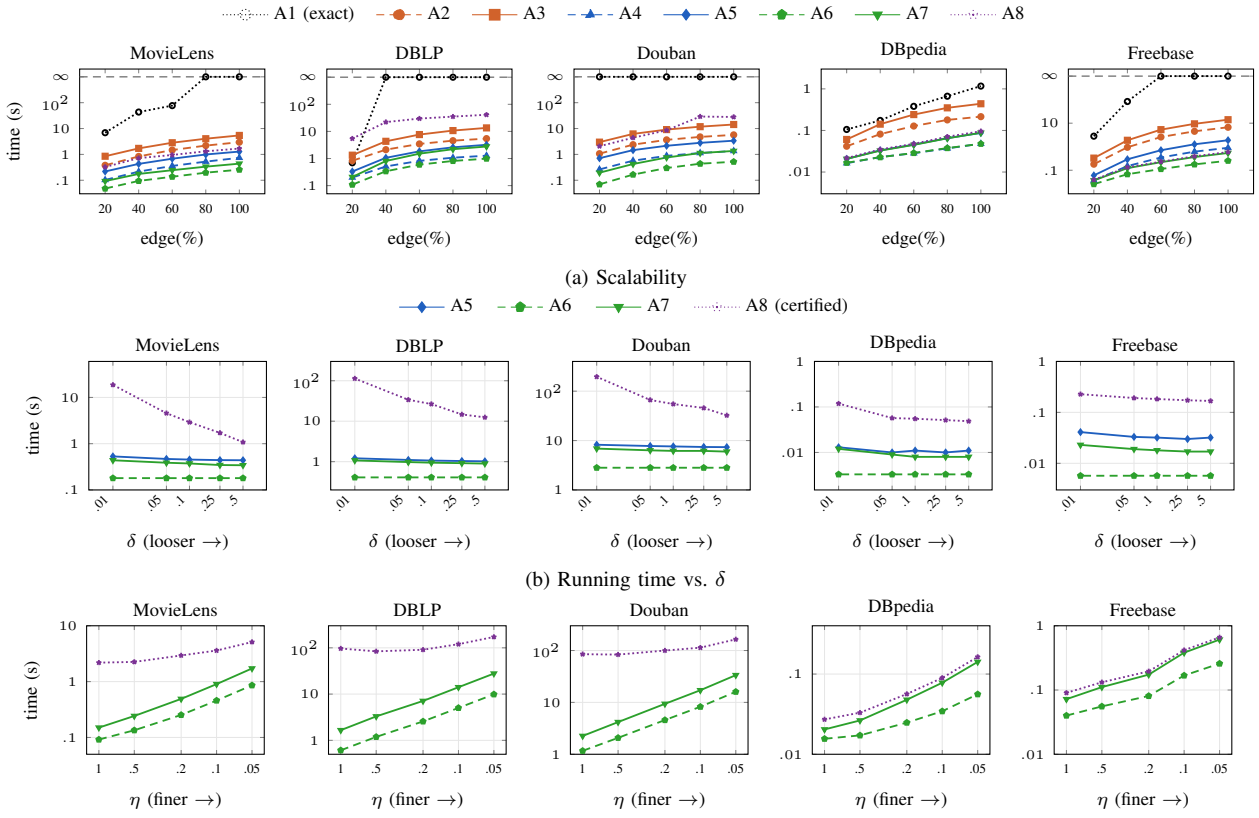


Fig. 6. Efficiency evaluation.

Parameters. We vary net granularity η, per-subproblem tolerance δ, and meta-path length i, with defaults 0.1, 0.05, 3 (4)[4], respectively.

B. Effectiveness

Except for A1, which supplies the reference optimum ρ^*, only results finishing within the cut-off time are shown.

Recovered density. Figures 4(a) and 4(b) report the density actually recovered, ρ/ρ^*, as η and δ vary. The first observation is that the near-optimal variants recover the optimum: on the core datasets A7 reaches $\rho/\rho^* \geq 0.999$ at the default $\eta{=}0.1$ (within $1.5{\times}10^{-3}$), and on the breadth datasets the near-optimal solver returns the exact optimum at every setting. The second observation is that the accuracy is insensitive to both parameters: A7 varies by about 1% across η and under 0.2% across δ. The reason is the quadratic net loss of Theorem 2, whose gap stays tiny even at a coarse η; hence coverage and quality decouple and a coarse (η, δ) incurs no loss. A faint ordering A3 $\gtrsim$ A5 $\gtrsim$ A7 also appears across δ: the un-reduced first-order net represents more weight sets, so its best cell lands slightly closer to the optimum, but the gap ($\approx 0.2\%$) is within the $(1{+}\eta)$ tolerance.

Achieving vs. certifying near-optimality (A6/A7 vs. A8). Whether a subproblem is peeled to the fixed $1/i$ cap (A6), to the heuristic near-optimal stop (A7), or to the worst-case $(1{-}\delta)$ certificate (A8), A7 and A8 recover the same density and A6 is within 0.4% of them (Figure 4(a)). The reason is that on real inputs the peel reaches a near-optimal subgraph within a few passes, so certifying it only *proves* the result rather than changing it. Consequently, certification yields a guarantee rather than accuracy: A6 and A7 achieve near-optimal density but cannot certify it on a given input, whereas A8's end-to-end $(1{-}\delta)/(1{+}\eta)$ bound, secured by its certificate, holds whatever the data (§VI-C shows an input where the uncertified $1/i$ peel falls short). Therefore, A6/A7 is preferable when the input is trusted and A8 when a worst-case guarantee is required.

Case study: drug repurposing. We run the motivating compound–gene–disease repurposing query of §I directly on HETIONET [2] (47K nodes, 2.25M edges), where, as discussed before, a densest $\langle \mathsf{C}, \mathsf{G}, \mathsf{D} \rangle$ block can serve as a repurposing shortlist. To make the result displayable, we filter out a few generic high-degree "housekeeping" hub genes (the analogue of stop-words); the query then returns a coherent module: a family of anti-inflammatory drugs, the shared enzymes they target (COX-2 and two relatives, the targets of aspirin and ibuprofen), and the inflammatory diseases those enzymes drive (Figure 5). The result can be checked against held-out labels: the query reads only the BINDS/ASSOCIATES edges but a treatment relation, yet five of the drug–disease pairs it implies (e.g. sulfasalazine–rheumatoid arthritis) are *approved* indications stored in a *separate, held-out* TREATS edge type, so the

[4] 3 on the core datasets, 4 on the breadth datasets

dense structure recovers approved indications it was not shown before. The remaining implied pairs (the same enzymes to three cancers, carrying *no* TREATS edge) match a documented but still-investigational therapy, plausible novel leads for a repurposing pipeline. The exact algorithm A1 confirms that the returned subgraph is optimal.

C. Efficiency

Scalability. Following the subgraph-generation of [7], for each dataset we retain fixed percentages (20–100%) of the edges (10 sets for each percentage) and run the algorithms on the induced subgraphs (Figure 6(a)); i is set to 3 on the core datasets and to 4 on the breadth datasets. The first observation is that only the localized-net variants A6, A7, and A8 stay nearly flat up to the complete dataset. The reason is that their net is n-independent and each pass keeps only $\mathcal{O}(|E|)$ state without materializing the $\mathcal{O}(\prod_j |V_j|)$ instances, so their cost grows only with the graph. In contrast, removing the reductions degrades scalability: the second-order net without localization (A4/A5), and far more so the un-localized first-order nets (A2/A3), grow sharply with the graph. On FREEBASE, A3 reaches 14 s against A7's 0.5 s, a 26× difference. A1, solving a flow subproblem per surviving iRM-set, is the least scalable, timing out on every core dataset and on FREEBASE. The results verify that both reductions, the second-order net and the Δ-box localization, are critical for scalability; we therefore drop A1–A4 below.

Achieving vs. certifying near-optimality (A7 vs. A8). Figure 6(b) varies the per-subproblem tolerance δ. The first observation is that *achieving* near-optimal density is inexpensive and δ-insensitive: the heuristic peel (A5, A7) costs under 1.5× as δ tightens by 50×, and the $1/i$ peel (A6) is δ-independent, both finishing within a few seconds. In contrast, *certifying* near-optimality (A8) is far more costly, and its cost is data-dependent. The reason is that the certified time is governed by the number of *near-tied* net cells: on MOVIELENS and the breadth datasets a single dominant cell lets the incumbent prune the rest, so certification stays within 40s across all δ; on DBLP and DOUBAN, many cells carry near-tied density and each must be certified to convergence, so the running time increases sharply with small δ.

Varying net granularity η. Figure 6(c) varies η on the full graph. The running time rises smoothly as η refines, matching the predicted $\mathcal{O}((\min\{\log n, \log \Delta\}/\sqrt{\eta})^{i-1})$ cell count, with A7 a roughly constant factor above the $1/i$ peel A6. The reason is the per-cell solver overhead, which is the only difference between the two. On the core datasets A8 rises even more slowly, 1.8–2.3× under 20× cell growth, as its time goes to the few near-tied cells, whose number barely changes with η.

Varying meta-path length i. Figure 7 varies i from 3 up to 9 on the breadth HINs, averaging over the query graphs at each length. The first observation is that only the localized variants remain tractable, their running time tracking the net's $\mathcal{O}((\min\{\log n, \log \Delta\}/\sqrt{\eta})^{i-1})$ growth, the one bottleneck the reductions cannot remove. The second observation is that the uncertified A6 and A7 finish every length (A7 within 34 s), whereas the certified A8 runs 4–9× slower when i is large.

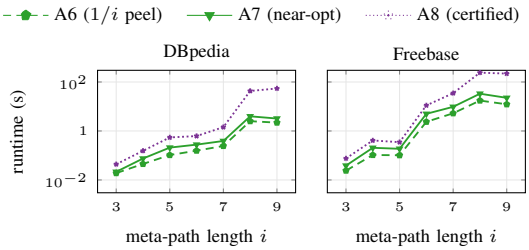


Fig. 7. Varying i on the breadth datasets.

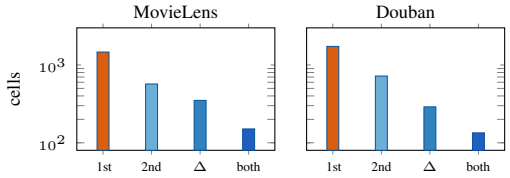


Fig. 8. Localization

This is consistent with the achieve-vs-certify gap above, now amplified by the $i-1$ exponent in the net.

Effect of localization. Figure 8 reports the net cell count through the following methods: 1) the first-order net (A2), 2) the second-order net (§V-B), 3) the Δ-box localization (§V-D), and both (2 plus 3). both composes to an ≈10× smaller net on the core datasets, which explains the fast running times of A6, A7, and A8 observed above.

IX. RELATED WORK

Densest subgraph and its variants. The densest subgraph, which maximizes $|E(S)|/|S|$, is exact via parametric max-flow [21] and 1/2-approximable by greedy peeling [12]; its variants span directed/bipartite [13], density-friendly [22], k-clique [23], [24], multilayer [6], anchored [25], and locally densest clique/triangle [26]–[28] density, recently consolidated in a unified analysis [29] and a broad survey [30]. DPSS is their i-partite meta-path generalization.

Near-optimal supermodular solvers. A fast line recasts the objective as a *supermodular* ratio solved to $(1-\varepsilon)$: Greedy++ [18], Super-Greedy++ [19], scalable decomposition [20] and its lexicographically-optimal-base Frank–Wolfe view [31], coordinate descent [17], and, for the directed case, convex programming [14], scalable discovery [32], and accelerated coordinate descent [15]. All are one-dimensional thanks to the scalar ratio, at $i = 2$, which they scan at pitch ε ($\mathcal{O}(\log n/\varepsilon)$ guesses [14]); our second-order net needs only $\mathcal{O}(\log n/\sqrt{\eta})$ (§V-B), and for $i > 2$, where the weights lose their total order (§III), we lift them via the η net based cover.

Dense subgraphs over HINs. HINs model typed, multi-relational data queried through meta-paths [1], [3], [33]; cohesive-structure discovery there centers on query-anchored *community search* [8]–[10], [34], with recent methods even avoiding meta-path *materialization* [35], akin to our instance-free peeling. The densest $\mathcal{P}$-partite subgraph [7] optimizes one parameter-free density globally; a recent exact variant [16]

adds box-level pruning of iRM-sets to the same flow-based search. We improve on both from the approximation side: a polylogarithmic cover replaces the iRM-set search, and each cell is solved by an instance-free peeling based algorithm.

X. Conclusion

In this paper, we devise approximation algorithms for the DPSS problem. We show that the exhaustive weight-set enumeration behind prior DPSS methods is unnecessary for approximation. The feasible iRM-sets fill a bounded $(i-1)$-dimensional polytope that an η-net covers with polylogarithmically many representatives. Each weighted supermodular densest-subgraph instance is then solved near-optimally and instance-free, with incumbent-driven pruning discarding most before applying any solver. Overall, our algorithm achieves a $\frac{1-\delta}{1+\eta}$ guarantee at polylogarithmic cost. Extensive experimental studies on real datasets justify the effectiveness and efficiency of our proposed methods.

References

[1] C. Shi, Y. Li, J. Zhang, Y. Sun, and P. S. Yu, "A survey of heterogeneous information network analysis," *IEEE Transactions on Knowledge and Data Engineering*, vol. 29, no. 1, pp. 17–37, 2017.

[2] D. S. Himmelstein, A. Lizee, C. Hessler, L. Brueggeman, S. L. Chen, D. Hadley, A. Green, P. Khankhanian, and S. E. Baranzini, "Systematic integration of biomedical knowledge prioritizes drugs for repurposing," *eLife*, vol. 6, p. e26726, 2017.

[3] Y. Sun, J. Han, X. Yan, P. S. Yu, and T. Wu, "PathSim: Meta path-based top-k similarity search in heterogeneous information networks," *Proceedings of the VLDB Endowment*, vol. 4, no. 11, pp. 992–1003, 2011.

[4] T. Grünert, S. Irnich, H.-J. Zimmermann, M. Schneider, and B. Wulfhorst, "Finding all k-cliques in k-partite graphs, an application in textile engineering," *Computers & Operations Research*, vol. 29, no. 1, pp. 13–31, 2002.

[5] J. Pei, D. Jiang, and A. Zhang, "On mining cross-graph quasi-cliques," in *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*. ACM, 2005, pp. 228–238.

[6] E. Galimberti, F. Bonchi, and F. Gullo, "Core decomposition and densest subgraph in multilayer networks," in *Proceedings of the ACM International Conference on Information and Knowledge Management (CIKM)*. ACM, 2017, pp. 1807–1816.

[7] L. Chen, C. Liu, R. Zhou, K. Liao, J. Xu, and J. Li, "Densest multipartite subgraph search in heterogeneous information networks," *Proceedings of the VLDB Endowment*, vol. 17, no. 4, pp. 699–711, 2023.

[8] X. Jian, Y. Wang, and L. Chen, "Effective and efficient relational community detection and search in large dynamic heterogeneous information networks," *Proceedings of the VLDB Endowment*, vol. 13, no. 10, pp. 1723–1736, 2020.

[9] Y. Zhou, Y. Fang, W. Luo, and Y. Ye, "Influential community search over large heterogeneous information networks," *Proceedings of the VLDB Endowment*, vol. 16, no. 8, pp. 2047–2060, 2023.

[10] Y. Fang, Y. Yang, W. Zhang, X. Lin, and X. Cao, "Effective and efficient community search over large heterogeneous information networks," *Proceedings of the VLDB Endowment*, vol. 13, no. 6, pp. 854–867, 2020.

[11] R. Dagher, M. Cohen, G. Williams, M. Rothmann, J. Gobburu, G. Robbie, A. Rahman, G. Chen, A. Staten, D. Griebel, and R. Pazdur, "Approval summary: Imatinib mesylate in the treatment of metastatic and/or unresectable malignant gastrointestinal stromal tumors," *Clinical Cancer Research*, vol. 8, no. 10, pp. 3034–3038, 2002.

[12] M. Charikar, "Greedy approximation algorithms for finding dense components in a graph," in *Proceedings of the International Workshop on Approximation Algorithms for Combinatorial Optimization (APPROX)*, ser. Lecture Notes in Computer Science, vol. 1913. Springer, 2000, pp. 84–95.

[13] S. Khuller and B. Saha, "On finding dense subgraphs," in *Proceedings of the International Colloquium on Automata, Languages and Programming (ICALP)*, ser. Lecture Notes in Computer Science, vol. 5555. Springer, 2009, pp. 597–608.

[14] C. Ma, Y. Fang, R. Cheng, L. V. S. Lakshmanan, and X. Han, "A convex-programming approach for efficient directed densest subgraph discovery," in *Proceedings of the ACM International Conference on Management of Data (SIGMOD)*. ACM, 2022, pp. 845–859.

[15] L. Liang, Y. Zhou, and Y. Fang, "Accelerated coordinate descent for directed densest subgraph discovery," in *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*. ACM, 2026, pp. 783–794.

[16] J. Xie, J. Yang, K. Zhao, and J. X. Yu, "Scalable exact densest $\mathcal{P}$-partite subgraph search in heterogeneous information networks," arXiv:2608.03061, 2026. [Online]. Available: https://arxiv.org/abs/2608.03061

[17] T. D. Nguyen and A. Ene, "Multiplicative weights update, area convexity and random coordinate descent for densest subgraph problems," in *Proceedings of the International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 235. PMLR, 2024, pp. 37 683–37 706.

[18] D. Boob, Y. Gao, R. Peng, S. Sawlani, C. E. Tsourakakis, D. Wang, and J. Wang, "Flowless: Extracting densest subgraphs without flow computations," in *Proceedings of the International Conference on World Wide Web (WWW)*. ACM / IW3C2, 2020, pp. 573–583.

[19] C. Chekuri, K. Quanrud, and M. R. Torres, "Densest subgraph: Supermodularity, iterative peeling, and flow," in *Proceedings of the ACM-SIAM Symposium on Discrete Algorithms (SODA)*. SIAM, 2022, pp. 1531–1555.

[20] E. Harb, K. Quanrud, and C. Chekuri, "Faster and scalable algorithms for densest subgraph and decomposition," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022, pp. 26 966–26 979.

[21] A. V. Goldberg, "Finding a maximum density subgraph," EECS Department, University of California, Berkeley, Tech. Rep. UCB/CSD-84-171, 1984.

[22] M. Danisch, T.-H. H. Chan, and M. Sozio, "Large scale density-friendly graph decomposition via convex programming," in *Proceedings of the International Conference on World Wide Web (WWW)*. ACM, 2017, pp. 233–242.

[23] C. E. Tsourakakis, "The k-clique densest subgraph problem," in *Proceedings of the International Conference on World Wide Web (WWW)*. ACM, 2015, pp. 1122–1132.

[24] Y. He, K. Wang, W. Zhang, X. Lin, and Y. Zhang, "Scaling up k-clique densest subgraph detection," *Proceedings of the ACM on Management of Data*, vol. 1, no. 1, pp. 1–26, 2023.

[25] Y. Dai, M. Qiao, and L. Chang, "Anchored densest subgraph," in *Proceedings of the ACM International Conference on Management of Data (SIGMOD)*. ACM, 2022, pp. 1200–1213.

[26] X. Xu, H. Liu, X. Lv, Y. Wang, and D. Li, "An efficient and exact algorithm for locally h-clique densest subgraph discovery," *Proceedings of the ACM on Management of Data*, vol. 2, no. 6, pp. 1–26, 2024.

[27] Y. Yang, C. Ma, R. Cheng, L. V. S. Lakshmanan, and X. Han, "Finding locally densest subgraphs: Convex programming with edge and triangle density," *The VLDB Journal*, vol. 35, no. 1, p. 7, 2026.

[28] L. Chen, C. Liu, K. Liao, J. Li, and R. Zhou, "Contextual community search over large social networks," in *2019 IEEE 35th International Conference on Data Engineering (ICDE)*, 2019, pp. 88–99.

[29] Y. Zhou, Q. Guo, Y. Yang, Y. Fang, C. Ma, and L. V. S. Lakshmanan, "In-depth analysis of densest subgraph discovery in a unified framework," *Proceedings of the VLDB Endowment*, vol. 18, no. 4, pp. 1131–1144, 2024.

[30] T. Lanciano, A. Miyauchi, A. Fazzone, and F. Bonchi, "A survey on the densest subgraph problem and its variants," *ACM Computing Surveys*, vol. 56, no. 8, pp. 208:1–208:40, 2024.

[31] E. Harb, K. Quanrud, and C. Chekuri, "Convergence to lexicographically optimal base in a (contra)polymatroid and applications to densest subgraph and tree packing," in *Proceedings of the Annual European Symposium on Algorithms (ESA)*, 2023, pp. 56:1–56:17.

[32] Y. Zhou, L. Liang, and Y. Fang, "Efficient and scalable directed densest subgraph discovery," *Proceedings of the ACM on Management of Data*, vol. 3, no. 6, pp. 1–27, 2025.

[33] Y. Sun and J. Han, *Mining Heterogeneous Information Networks: Principles and Methodologies*, ser. Synthesis Lectures on Data Mining and Knowledge Discovery. Morgan and Claypool Publishers, 2012.

[34] Y. Fang, X. Huang, L. Qin, Y. Zhang, W. Zhang, R. Cheng, and X. Lin, "A survey of community search over big graphs," *The VLDB Journal*, vol. 29, no. 1, pp. 353–392, 2020.

[35] J. Jiang, S. Yao, Y. Chen, B. He, Y. Niu, Y. Li, S. Sun, and Y. Liu, "Community detection in heterogeneous information networks without materialization," *Proceedings of the ACM on Management of Data*, vol. 3, no. 3, pp. 1–27, 2025.

APPENDIX A
FULL PROOF OF THE ITERATIVE PEELING SOLVER

Throughout, $\mathbf{m} = (m_1, \dots, m_i)$ is a fixed iRM-set of per-type weights; $f(S) = |\mathcal{F}(S)|$ counts the instances of the length-i meta-path $\mathcal{P}$ within $S \subseteq U$, the position-expanded ground set (f is normalized, monotone, and supermodular by Lemma 6); $m_v := m_{\phi'(v)} > 0$ where $\phi'(v) \in [i]$ is v's position in $\mathcal{P}$, $w_{\mathbf{m}}(S) = \sum_{v \in S} m_v$, $h_{\mathbf{m}}(S) = f(S)/w_{\mathbf{m}}(S)$ for $\emptyset \neq S \subseteq U$, $h^*_{\mathbf{m}} = \max_{\emptyset \neq S} h_{\mathbf{m}}(S)$, $P(v, H)$ is the number of instances through v in a subgraph H, $\Delta_j = \max_{v \in V(A_j)} P(v, G)$ ($V(A_j)$ the vertices of type A_j), and $\Delta = \max_j \Delta_j$, the maximum per-vertex instance count. Deleting v from S destroys exactly the instances through v, so, with $G(S)$ the subgraph induced by S, $P(v, G(S)) = f(S) - f(S \setminus v) = f(v \,|\, S \setminus v)$. A vector $x \in \mathbb{R}^U$ has one coordinate x_v per element $v \in U$, and we write $x(S) = \sum_{v \in S} x_v$; the (contra)polymatroid base polytope of f is

$$B(f) = \{x \geq 0 : x(S) \geq f(S)\ \forall S \subseteq U,\ x(U) = f(U)\}.$$

For every $x \in B(f)$ and $v \in U$, $f(\{v\}) \leq x_v = x(U) - x(U - v) \leq f(U) - f(U - v) = f(v \,|\, U - v) = P(v, G) \leq \Delta$, so $B(f) \subseteq [0, \Delta]^U$ is compact and convex. The *weighted width* is $\Delta_{\mathbf{m}} = \max_{v \in U} P(v, G)/m_v = \max_j \Delta_j/m_j$ (m_v is constant on layers), computable by the same two sweeps that build the Δ-box (Observations 1–2).

The algorithm in full. ITERATIVE PEELING$(f, \mathbf{m}, T)$: set $\sigma \equiv 0$; for $t = 1, \dots, T$: set $S \leftarrow U$ and, while $S \neq \emptyset$, record S as a candidate family, then delete

$$u = \arg\min_{v \in S} \frac{\sigma(v) + P(v, G(S))}{m_v},$$
$$g_t(u) := P(u, G(S)), \qquad \sigma(u) \mathrel{+}= g_t(u).$$

Return the densest candidate family $\hat{S}$ over all prefixes (the surviving sets just before each deletion) of all passes. At $T{=}1$, $\sigma \equiv 0$ and the key is $P(v, G(S))/m_v$, exactly the reweighted peel of Theorem 1.

Lemma A.1 (Greedy bases; exact linear minimization)**.** *Let f be normalized, monotone, and supermodular on U. For an ordering $\pi = (\pi_1, \dots, \pi_{|U|})$ of U with prefixes $P_k = \{\pi_1, \dots, \pi_k\}$ define the* greedy base *g^π by $g^\pi_{\pi_k} := f(\pi_k \,|\, P_{k-1})$. Then: (a) $g^\pi \in B(f)$ for every ordering π; (b) for every $z \in \mathbb{R}^U_{\geq 0}$, if π sorts z decreasingly ($z_{\pi_1} \geq \cdots \geq z_{\pi_{|U|}}$, ties arbitrary), then*

$$\min_{x \in B(f)} \langle z, x \rangle = \langle z, g^\pi \rangle = \int_0^\infty f\big(\{v : z_v \geq \tau\}\big)\, d\tau =: \hat{f}(z),$$

the Lovász extension of f at z.

Proof. (a) *Nonnegativity:* $g^\pi_{\pi_k} = f(\pi_k \,|\, P_{k-1}) \geq f(\{\pi_k\}) \geq 0$, since supermodular marginals are nondecreasing in the context set and f is monotone. *Tightness:* $g^\pi(U) = \sum_k \big(f(P_k) - f(P_{k-1})\big) = f(U)$ telescopes. *Covering:* fix $S \subseteq U$ and enumerate it in π-order as $s_1, \dots, s_r$ with $s_j = \pi_{k_j}$; then $\{s_1, \dots, s_{j-1}\} \subseteq P_{k_j - 1}$, so by monotone marginals

$$f(S) = \sum_{j=1}^r f\big(s_j \,\big|\, \{s_1, \dots, s_{j-1}\}\big) \leq \sum_{j=1}^r f\big(s_j \,\big|\, P_{k_j - 1}\big) = g^\pi(S).$$

(b) Set $z_{\pi_{|U|+1}} := 0$. Abel summation gives, for any x, $\langle z, x \rangle = \sum_k \big(z_{\pi_k} - z_{\pi_{k+1}}\big) x(P_k)$, with all coefficients ≥ 0 by the sorting. For $x \in B(f)$, $x(P_k) \geq f(P_k)$ for $k < |U|$ and $x(P_{|U|}) = f(U)$; the greedy base attains $g^\pi(P_k) = f(P_k)$ for all k. Hence $\langle z, x \rangle \geq \sum_k (z_{\pi_k} - z_{\pi_{k+1}}) f(P_k) = \langle z, g^\pi \rangle$. Finally $\{v : z_v \geq \tau\} = P_k$ for $\tau \in (z_{\pi_{k+1}}, z_{\pi_k}]$, an interval of length $z_{\pi_k} - z_{\pi_{k+1}}$, so the sum equals the integral defining $\hat{f}(z)$. □

Lemma A.2 (Weighted quasi-min-max)**.** *With $f, \mathbf{m}$ as above and $f \not\equiv 0$,*

$$h^*_{\mathbf{m}} = \max_{\emptyset \neq S \subseteq U} \frac{f(S)}{w_{\mathbf{m}}(S)} = \min_{x \in B(f)} \max_{v \in U} \frac{x_v}{m_v}.$$

Proof. Weak direction ($\leq$). For $x \in B(f)$ and $\emptyset \neq S$: $f(S) \leq x(S) = \sum_{v \in S} \frac{x_v}{m_v} m_v \leq \big(\max_{v \in S} \frac{x_v}{m_v}\big) w_{\mathbf{m}}(S)$, so $h_{\mathbf{m}}(S) \leq \max_{v \in U} x_v/m_v$ for every S and every $x \in B(f)$.

Strong direction ($\geq$). Let $Z := \{z \in \mathbb{R}^U_{\geq 0} : \sum_v m_v z_v = 1\}$, compact and convex since every $m_v > 0$. *Step 1:* $h^*_{\mathbf{m}} = \max_{z \in Z} \hat{f}(z)$. For "$\leq$": $z := \mathbf{1}_{S^*}/w_{\mathbf{m}}(S^*) \in Z$ for a maximizer S^*, and $\hat{f}(z) = f(S^*)/w_{\mathbf{m}}(S^*) = h^*_{\mathbf{m}}$ by the integral formula of Lemma A.1(b). For "$\geq$": take any $z \in Z$ sorted decreasingly by π with prefixes P_k and coefficients $\alpha_k := z_{\pi_k} - z_{\pi_{k+1}} \geq 0$; then $\hat{f}(z) = \sum_k \alpha_k f(P_k)$ and $1 = \sum_v m_v z_v = \sum_k \alpha_k w_{\mathbf{m}}(P_k)$, so $f(P_k) \leq h^*_{\mathbf{m}} w_{\mathbf{m}}(P_k)$ (each $P_k \neq \emptyset$) gives $\hat{f}(z) \leq h^*_{\mathbf{m}}$. *Step 2: exchange min and max.* By Lemma A.1(b), $\hat{f}(z) = \min_{x \in B(f)} \langle z, x \rangle$ on $Z \subseteq \mathbb{R}^U_{\geq 0}$. The payoff $\langle z, x \rangle$ is bilinear and both Z and $B(f)$ are compact convex sets, so von Neumann's minimax theorem gives

$$h^*_{\mathbf{m}} = \max_{z \in Z} \min_{x \in B(f)} \langle z, x \rangle = \min_{x \in B(f)} \max_{z \in Z} \langle z, x \rangle = \min_{x \in B(f)} \max_{v \in U} \frac{x_v}{m_v},$$

the last step because a linear functional over the weighted simplex Z is maximized at an extreme point $z = \mathbf{e}_v/m_v$. □

Lemma A.3 (Per-pass base membership)**.** *Consider one pass of* ITERATIVE PEELING *(with* any *deletion key), deleting $u_1, u_2, \dots, u_{|U|}$ in order, where $S_k = \{u_k, \dots, u_{|U|}\}$ is the surviving set before the k-th deletion, and charging $g(u_k) = P(u_k, G(S_k)) = f(u_k \,|\, S_k \setminus u_k)$. Then $g = g^\pi$ for the* reverse *deletion order $\pi = (u_{|U|}, \dots, u_1)$; hence $g \in B(f)$ by Lemma A.1(a). Consequently, after t passes with charges $g_1, \dots, g_t$ and accumulated load $\sigma_t := \sum_{s \leq t} g_s$, the averaged load $\bar{x}_t := \sigma_t/t$ lies in $B(f)$ by convexity.*

Proof. Under $\pi = (u_{|U|}, \dots, u_1)$ the element at position j is $u_{|U|-j+1}$ and the prefix is $P_{j-1} = \{u_{|U|}, \dots, u_{|U|-j+2}\} = S_{|U|-j+1} \setminus u_{|U|-j+1}$; its greedy marginal is therefore $f(u_{|U|-j+1} \,|\, S_{|U|-j+1} \setminus u_{|U|-j+1}) = g(u_{|U|-j+1})$, coordinate-wise identical to the pass's charge. □

The membership in Lemma A.3 holds for *any* deletion key: every deletion order emits some greedy base; the key only selects which one. No exact linear-minimization claim is made for the load-augmented key: the reverse deletion order need not sort either the pre-pass or the post-pass loads, which is why the convergence of Greedy++ was long a conjecture [18], resolved by a tailored analysis [19].

Proof of Proposition 1. Part 1: Sound reporting via weak duality (steps (i)–(ii) of the main-paper proof sketch). By Lemma A.3, after any t passes $\bar{x}_t = \sigma_t/t \in B(f)$, so by the weak direction of Lemma A.2,

$$h^*_{\mathbf{m}} \le \max_{v\in U} \frac{\bar{x}_t(v)}{m_v} \quad \text{and} \quad h_{\mathbf{m}}(\hat{S}) \ge \max_{v} \frac{\bar{x}_t(v)}{m_v} - \hat{g}_t, \quad (3)$$

where $\hat{g}_t := \max_v \bar{x}_t(v)/m_v - \max_{\text{prefixes } S} h_{\mathbf{m}}(S)$ is the nonnegative, computable duality gap after t passes. Thus $\hat{g}_t \le \delta\, h^*_{\mathbf{m}}$ certifies $h_{\mathbf{m}}(\hat{S}) \ge (1-\delta)h^*_{\mathbf{m}}$, unconditionally.

Part 2: The gap closes at the stated rate (sketch step (iii)). It remains to bound $\min_{t\le T} \hat{g}_t$. We separate what is proved here from what is imported, since only one step of the argument relies on an external result.

Correctness is self-contained. By (3), whenever the computable gap obeys $\hat{g}_t \le \delta\, h^*_{\mathbf{m}}$, the reported family satisfies $h_{\mathbf{m}}(\hat{S}) \ge (1-\delta)h^*_{\mathbf{m}}$, and this uses *only* weak duality (Lemma A.2) and averaged-base membership (Lemma A.3), both established above for the per-type-weighted objective. Thus every family the algorithm reports is certified without appeal to any external rate (Remark A.1). What is *imported* from [19] is solely the a priori guarantee that this gap test is met within the stated number of passes; we now show that this rate transfers to the weighted objective once every per-element quantity is read per unit weight, rather than reproving it.

The imported rate. We invoke the analysis behind Theorem 1.3 of [19]: for a normalized, monotone, supermodular F on n elements, SUPER-GREEDY++ (peel by $\sigma(v) + F(v \mid S\setminus v)$, charge the marginal, report the best prefix) attains $\min_{t\le T} \hat{g}^{\text{card}}_t \le \varepsilon\lambda^*$ once $T \ge C\,\Delta_F \ln n/(\varepsilon^2\lambda^*)$, where $\Delta_F = \max_v F(v \mid V-v)$, $\lambda^* = \max_{\emptyset\ne S} F(S)/|S|$, C is an absolute constant, and $\hat{g}^{\text{card}}_t$ is the cardinality-denominator gap; the theorem is stated there as a $(1-\varepsilon)$-approximation, and its proof establishes the duality-gap form quoted here. (This is the rate whose tightness [19] leave open; see Remark A.3.)

Reading the analysis per unit weight. Regard each element v as carrying its weight $m_v = m_{\phi'(v)} > 0$; every per-element quantity is then read per unit weight, under the correspondence

role in [19]	unweighted	weighted
per-element load	$\sigma(v)$	$\sigma(v)/m_v$
per-element marginal	$F(v \mid S\setminus v)$	$P(v, G(S))/m_v$
deletion key	$\sigma(v)+F(v \mid S\setminus v)$	$(\sigma(v)+P(v, G(S)))/m_v$
denominator	$\lvert S\rvert$	$w_{\mathbf{m}}(S) = \sum_{v\in S} m_v$
duality anchor	$\min_{x\in B(F)} \max_v x_v$	$\min_{x\in B(f)} \max_v x_v/m_v$
width	$\max_v F(v \mid V-v)$	$\max_v P(v, G)/m_v$
log factor	$\ln n$	$\ln \lvert U\rvert$

the anchor being λ^* (unweighted) and $h^*_{\mathbf{m}}$ (weighted, Lemma A.2), the width Δ_F and $\Delta_{\mathbf{m}}$. Under this correspondence the proof of [19] transfers step by step, its steps falling into two kinds.

Combinatorial steps are independent of the weights. The peeling order, the prefix accounting, and base membership never consult the weights. No matter which key chooses the vertex, a pass charges each deleted v its marginal $P(v, G(S)) = f(v \mid S\setminus v)$, so in reverse deletion order it emits a greedy base of f, and the averaged loads $\bar{x}_t = \sigma_t/t$ lie in $B(f)$; this is Lemma A.3, proved with no reference to $\mathbf{m}$. Which prefixes are charged, and in what order, is identical to the unweighted run; the weights enter only afterward, dividing the value attached to a coordinate.

Analytic steps are the same statement under a change of functional. By (3) our goal is to certify $\max_v \bar{x}_t(v)/m_v \to h^*_{\mathbf{m}}$, while [19] certify $\max_v \bar{x}_t(v) \to \lambda^*$. These are one statement with the linear functional $\max_v(\cdot)_v$ replaced by $\max_v(\cdot)_v/m_v$ and λ^* replaced by $h^*_{\mathbf{m}}$. Lemma A.2 shows the latter is again a min–max over the *same* base polytope $B(f)$, of the same combinatorial type (the Lovász extension of f paired against the weighted simplex $Z = \{z \ge 0 : \sum_v m_v z_v = 1\}$; Lemma A.2, Step 2). Hence every inequality in [19] relating $\max_v \bar{x}_t(v)$ to λ^* has an exact counterpart relating $\max_v \bar{x}_t(v)/m_v$ to $h^*_{\mathbf{m}}$, and the counterpart is valid because the three inputs those inequalities depend on are all shared and weight-free: the polytope $B(f)$, the greedy/linear-minimization identity (Lemma A.1), and base membership (Lemma A.3). Only two quantities update: the coordinate range under the new functional, $\Delta_F \to \Delta_{\mathbf{m}}$, and the logarithmic factor, $\ln n \to \ln |U|$.

Where the $\ln |U|$ *and the width enter.* Two ingredients of [19] fix these factors, and the rescaling touches each transparently. First, the regret term arises from an entropy over the max-player's pure strategies; those strategies are the $|U|$ coordinate directions $\mathbf{e}_v/m_v$ spanning Z, so their number is $|U|$ whatever the weights m_v; rescaling a weight rescales its direction but adds no strategy, which is exactly why the logarithm is $\ln |U|$ and never $\ln w_{\mathbf{m}}(U)$. Second, the per-round quantities the analysis bounds are the per-unit-weight values x_v/m_v, whose range is the width $\Delta_{\mathbf{m}}$; the weights enter the rate only by setting this range. The precise assembly of these two ingredients into the pass count $\mathcal{O}(\Delta_{\mathbf{m}} \ln |U|/(\delta^2 h^*_{\mathbf{m}}))$ (linear in the width and in $1/h^*_{\mathbf{m}}$, quadratic in $1/\delta$) is the tailored analysis of [19], the same argument now read per unit weight, and we import it as such; the $1/\delta^2$ exponent proper to peeling-type schemes is discussed in Remark A.3.

Conclusion. Combining,

$$\min_{t\le T} \hat{g}_t \le \delta\, h^*_{\mathbf{m}} \quad \text{once} \quad T = \mathcal{O}\big(\Delta_{\mathbf{m}} \ln |U|/(\delta^2 h^*_{\mathbf{m}})\big),$$

and (3) converts this into the claimed $(1-\delta)$ guarantee for the reported $\hat{S}$. We claim sufficiency of this T, not tightness: [19] pose the precise rate of SUPER-GREEDY++ as an open problem, and any sharpening transfers through the same correspondence untouched. Independently of the imported rate, the reported family is certified by (3) whenever the computable gap test is met (Remark A.1).

Part 3: Width on the localized net; the $T{=}1$ *case* (completing sketch step (iii)). Since m_v is constant on layers, $\Delta_{\mathbf{m}} = \max_j \Delta_j/m_j$, an identity in the paper's existing parameters.

The net solved is the Δ-box enlarged by one lattice step $s(\eta) = \sqrt{2\eta/e}$, at most 1 since $\eta \le 1$ (Definitions 1 and 2), so its points satisfy $\ln m_j \ge -\ln\Delta - s(\eta)$, hence $m_j \ge e^{-1}/\Delta$; with $\Delta_j \le \Delta$ this gives $\Delta_{\mathbf{m}} \le e\,\Delta^2 = \mathcal{O}(\Delta^2)$ uniformly over net cells (the cell-exact accounting at the optimal cell's representative is Remark A.2). Finally, at $T{=}1$, $\sigma \equiv 0$ and the key is $P(v, G(S))/m_v$: pass 1 coincides exactly with the reweighted peel of Theorem 1, whose $\frac{1}{i}$ guarantee transfers to $h_{\mathbf{m}}$ since the weighted density of §V-C equals $i\,h_{\mathbf{m}}$; the load carried across passes is what lifts it to $(1-\delta)$. □

Remark A.1 (Computable certificate; early stopping)**.** *After any t passes, both sides of* (3) *are computable in $\mathcal{O}(|U|)$ from σ_t and the incumbent prefix $\hat{S}_t$: if $\max_{v\in U}\sigma_t(v)/(t\,m_v) \le h_{\mathbf{m}}(\hat{S}_t)/(1-\delta)$, then $h_{\mathbf{m}}(\hat{S}_t) \ge (1-\delta)h^*_{\mathbf{m}}$,* unconditionally *(weak duality plus Lemma A.3; no imported rate is involved). Proposition 1 guarantees the test is met within the stated T; in practice it is met far earlier (§VIII).*

Remark A.2 (Uniform pass budget: $T = \mathcal{O}(i\,\Delta \ln|U|/\delta^2)$)**.** *Theorem 5 invokes the $(1-\delta)$ guarantee* only at the optimal cell's representative $\mathbf{m}'$*; every other cell merely contributes a candidate family. At that cell the width-to-value ratio collapses to existing parameters. Let $\mathcal{V}^*$ be the optimal family, ρ^* its density, and $\mathbf{m}^*$ its own iRM-set; by Lemma 5, $m^*_j = \bar{P}_j/\rho^*$ with $1 \le \bar{P}_j \le \Delta_j$. At $\mathbf{m}^*$ itself both quantities are exact: $h^*_{\mathbf{m}^*} = \rho^*/i$ (attained at $\mathcal{V}^*$ by AM–GM tightness of the induced iRM-set, while Lemma 7 caps $h^*_{\mathbf{m}} \le \rho^*/i$ for every feasible $\mathbf{m}$) and $\Delta_{\mathbf{m}^*} = \max_j \Delta_j/m^*_j = \rho^* \max_j \Delta_j/\bar{P}_j$, whence $\Delta_{\mathbf{m}^*}/h^*_{\mathbf{m}^*} = i\max_j \Delta_j/\bar{P}_j$. The net representative inherits this up to constants: Lemma 3 gives $|\ln m'_j - \ln m^*_j| \le s(\eta) \le 1$ coordinatewise, so $\Delta_{\mathbf{m}'} \le e^{s(\eta)}\rho^* \max_j \Delta_j/\bar{P}_j$, and Lemma 8 gives $h^*_{\mathbf{m}'} \ge \rho^*/(i(1+\eta))$; the two occurrences of ρ^** cancel*:*

$$\frac{\Delta_{\mathbf{m}'}}{h^*_{\mathbf{m}'}} \le e^{s(\eta)}\, i(1+\eta) \max_{j\in[i]} \frac{\Delta_j}{\bar{P}_j} \le e\, i(1+\eta)\,\Delta,$$

using $\bar{P}_j \ge 1$ and $\Delta_j \le \Delta$. Hence, absorbing $1+\eta \le 2$ into the constant, the uniform per-cell budget $T = \mathcal{O}(i\,\Delta\ln|U|/\delta^2)$ meets Proposition 1's requirement at $\mathbf{m}'$, and Theorem 5 holds unchanged with a budget in the paper's existing parameters, with no $h^*_{\mathbf{m}}$. The mechanism is that at the optimal cell the same ρ^* that inflates the width (through $m^*_j = \bar{P}_j/\rho^*$) also inflates the value, leaving the ratio $\max_j \Delta_j/\bar{P}_j$ (the global maximum incidence over the optimum's per-layer average incidence), which is near 1 when the optimum contains no vertex of exceptionally high incidence. Two qualifications are in order. First, the Δ-only form is a pipeline *fact: as a standalone per-cell statement, Proposition 1 does need $h^*_{\mathbf{m}}$, since an arbitrary net cell can push $\Delta_{\mathbf{m}}/h^*_{\mathbf{m}}$ well beyond $i\Delta$; only at $\mathbf{m}'$ does the cancellation occur. Second, Δ is the incidence of the graph the peeler actually runs on, so the pruning of §VII shrinks the budget further; cells may also stop early by Remark A.1.*

Remark A.3 (On the $1/\delta^2$ exponent)**.** *$\mathcal{O}(1/\varepsilon)$ iteration counts in this line of work come from accelerated first-order methods [20], which maintain* per-edge load-sharing variables*; the meta-path analogue materializes one variable per (instance, position) pair, $\Theta(i\,|\mathcal{F}(U)|)$ state, exactly the blow-up that instance-freeness (C3, §VI-D) is designed to avoid, and that the $L(v)R(v)$ factorization (Observation 1) cannot compress, since those iterates are not constant on instance classes. Peeling-type schemes, whose state is $\mathcal{O}(|U|)$, currently pay $1/\delta^2$ [19].*

Appendix B
Deferred Proofs

Proof of Theorem 1. For self-completeness, we give an alternative proof. Let $h^*_{\mathbf{m}} = \max_{S\subseteq U} h_{\mathbf{m}}(S)$, attained with vertex set S°. This indicates two facts below. (a) Every $v \in S^\circ$ has $P(v, G(S^\circ)) \ge h^*_{\mathbf{m}} m_{\phi'(v)}$: otherwise deleting v removes $P(v, G(S^\circ)) < h^*_{\mathbf{m}} m_{\phi'(v)}$ from f and $m_{\phi'(v)}$ from $w_{\mathbf{m}}$, raising the ratio above $h^*_{\mathbf{m}}$. (b) For every S, $w_{\mathbf{m}}(S) = \sum_j m_j|S_j| \ge i(\prod_j m_j|S_j|)^{1/i} = i\,g(S)$ by AM–GM and $\prod_j m_j = 1$, so $\rho(S) \ge i\,h_{\mathbf{m}}(S)$, with equality when $\mathbf{m}$ is S's own iRM-set; hence $h^*_{\mathbf{m}} \ge \rho^*_{\mathbf{m}}/i$. Now consider the pass at the moment it deletes the first vertex v of S°, from the current $S \supseteq S^\circ$. This moment arrives before the peeling empties any layer of S. Deleting vertices only removes induced instances, so $P(v, G(S)) \ge P(v, G(S^\circ))$ and, by (a), v's key satisfies $P(v, G(S))/m_{\phi'(v)} \ge h^*_{\mathbf{m}}$; as v minimizes the key, $P(u, G(S)) \ge h^*_{\mathbf{m}} m_{\phi'(u)}$ for all $u \in S$. Summing over S (each instance is counted at its i endpoints) gives $i f(S) \ge h^*_{\mathbf{m}} w_{\mathbf{m}}(S)$, so by (b) $\rho(S) \ge i\,h_{\mathbf{m}}(S) \ge h^*_{\mathbf{m}} \ge \rho^*_{\mathbf{m}}/i$; the peeling keeps a result at least this dense. □

Proof of Lemma 4. The bound factors into a *weight* step and a *solver* step. Let S_v be the subgraph present when peeling deletes v, and $\mu_j = \max\{P(v, S_v) : v \in V^*_j\}$.

Weight step. Charge each instance of $\mathcal{F}(\mathcal{V}^*)$ to its first-deleted endpoint; that endpoint v is deleted while all of $\mathcal{V}^*$ is still present, so the instance is counted in $P(v, S_v)$. Hence $|\mathcal{F}(\mathcal{V}^*)| \le \sum_{v\in\mathcal{V}^*} P(v, S_v) \le \sum_j |V^*_j|\mu_j$, and dividing by $g(\mathcal{V}^*)$ with $|V^*_j| = g(\mathcal{V}^*)/m^*_j$ gives $\rho^* \le \sum_j \mu_j/m^*_j$.

Solver step. The deleted vertex minimizes $P(u, S_v)/m'_{\phi'(u)}$ over $u \in S_v$, so this value is at most the m'-weighted average $i|\mathcal{F}(S_v)|/w_{\mathbf{m}'}(S_v) \le \rho(S_v) \le \widehat{\gamma}$ (AM–GM, $w_{\mathbf{m}'} \ge ig$); thus $\mu_j \le \widehat{\gamma}\, m'_j$.

Splitting $\mu_j/m^*_j = (\mu_j/m'_j)\,e^{\theta_j}$ and using $\sum_j e^{\theta_j} \le i(1+\eta)$ (Theorem 2),

$$\rho^* \;\le\; \sum_j \frac{\mu_j}{m^*_j} \;\le\; \widehat{\gamma}\sum_j e^{\theta_j} \;\le\; i(1+\eta)\,\widehat{\gamma}. \qquad \square$$

Proof of Corollary 1. Since $\frac{1-\zeta/3}{1+\zeta/3} \ge 1-\zeta$, Theorem 5 gives the factor. The $1/\zeta$-degree is $\frac{i+3}{2}$: $(i-1)/2$ from the $\mathcal{O}\big((\log n/\sqrt{\zeta}\,)^{i-1}\big)$ cells and 2 from the solver's $\mathcal{O}(1/\delta^2)$ passes (Proposition 1 with $\delta = \zeta/3$); the uniform pass budget $T = \mathcal{O}(i\,\Delta\ln|U|/\delta^2)$ is polynomial in $|G|$ at fixed i. The degree grows with i, and the concrete end-to-end runtime is given in §VI-D. □

Proof of Lemma 9. Fix a maximizer S°. Since $f(S^\circ) = h^*_{\mathbf{m}} w_{\mathbf{m}}(S^\circ) > 0$, every layer of S° is nonempty, so $w_{\mathbf{m}}(S^\circ) - m_{\phi'(v)} > 0$ for every $v \in S^\circ$. *(i)* For $v \in S^\circ$, deleting v destroys exactly $P(v, G(S^\circ))$ instances, and optimality of S° over the competitor $S^\circ \setminus \{v\}$ gives $\frac{f(S^\circ) - P(v, G(S^\circ))}{w_{\mathbf{m}}(S^\circ) - m_{\phi'(v)}} \le$

$h^*_{\mathbf{m}}$; cross-multiplying yields $P(v, G(S^\circ)) \geq h^*_{\mathbf{m}} m_{\phi'(v)}$, and $P(v, G) \geq P(v, G(S^\circ))$ since $S^\circ \subseteq U$. *(ii)* By (i) every $v \in S^\circ$ clears the floor $\lambda m_{\phi'(v)}$, so S° survives each round (with $P(v, G(U_r)) \geq P(v, G(S^\circ))$ as $S^\circ \subseteq U_r$, the ground set after r rounds, by induction) and stays optimal over the shrinking ground set, leaving $h^*_{\mathbf{m}}$ unchanged. $\square$